

End-to-End Bandgap Reference Design: From Schematic to Layout Essentials

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Erlangen - 24/02/2026

Projects: BMBF -> FMD-QNC (16ME0831)

Key Takeaways from Yesterday



- 0 Introduction to the IHP Open PDK
- 0 Learning the basics in Xschem related to simulations
- 0 Basic Klayout maneuvering
- 0 First examples on physical verification

Questions?

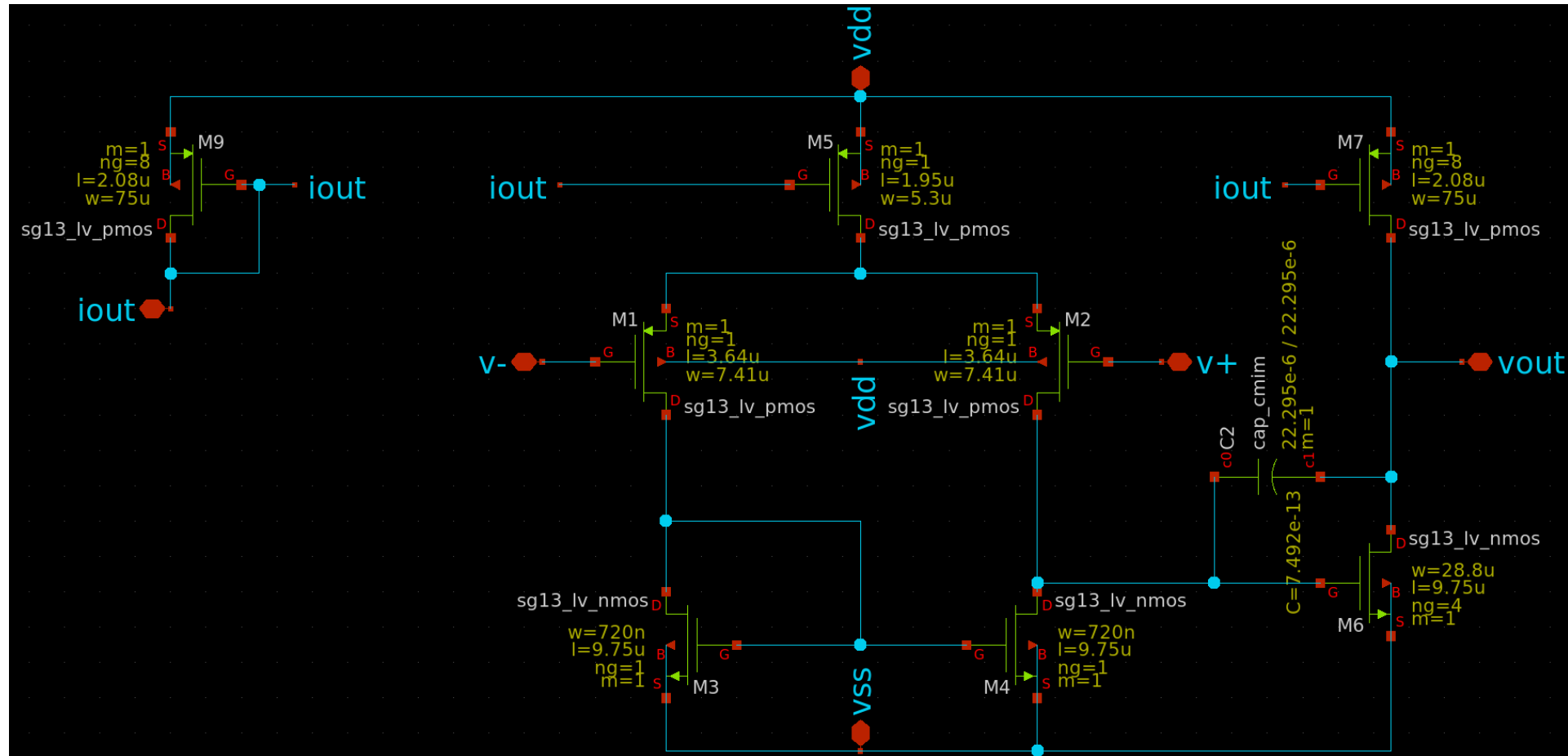
- 0 Was something from yesterday unclear?
- 0 Are we going to fast?
- 0 Do you need more time for the small exercise?

Overview Of Todays Plan



- 0 **OTA Design & Analysis:** Perform DC and AC analysis to evaluate key design metrics.
- 0 **Bandgap Voltage Reference:** Conduct DC, transient, and mismatch analysis using Monte Carlo sampling.
- 0 **From Design to Tape-Out:** Explore the final layout and discuss the steps required for tape-out in the open-source domain.
- 0 **Layout:** Practicing layout

OTA Design (Two Stage Miller Compensated)



Design Methodology In OS (Gm/ID)



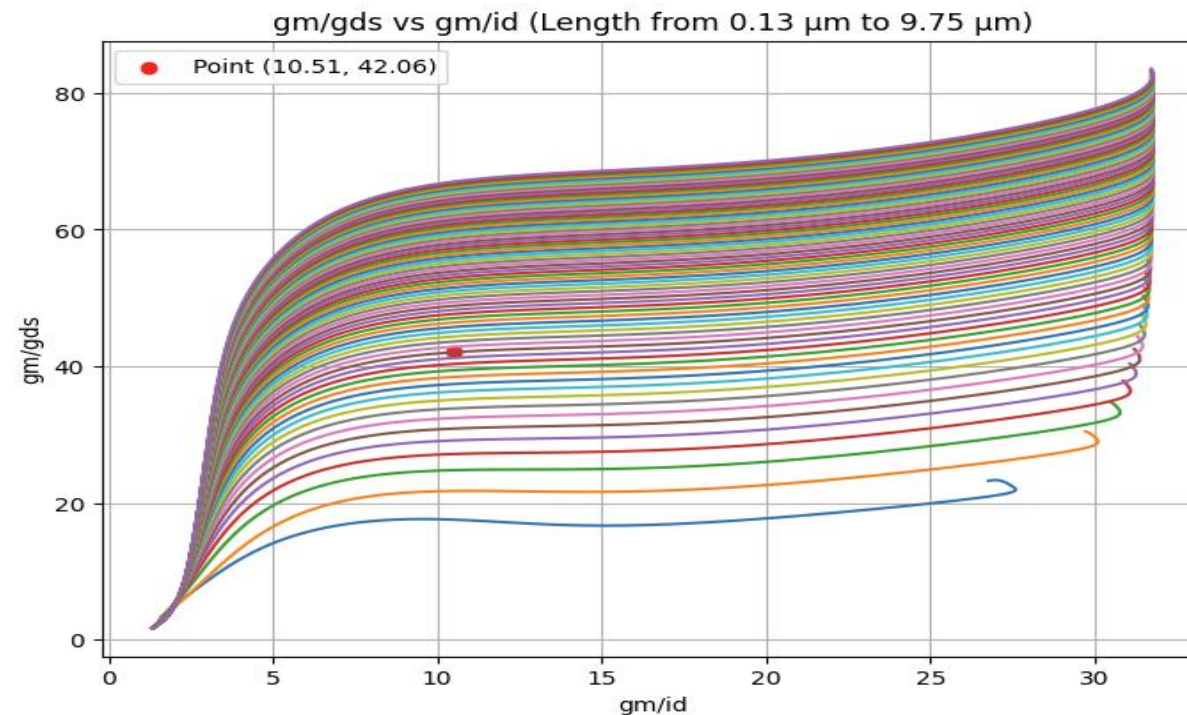
Select Lengt... Show All ▼

☒ Select gm/id (uncheck for gm/...

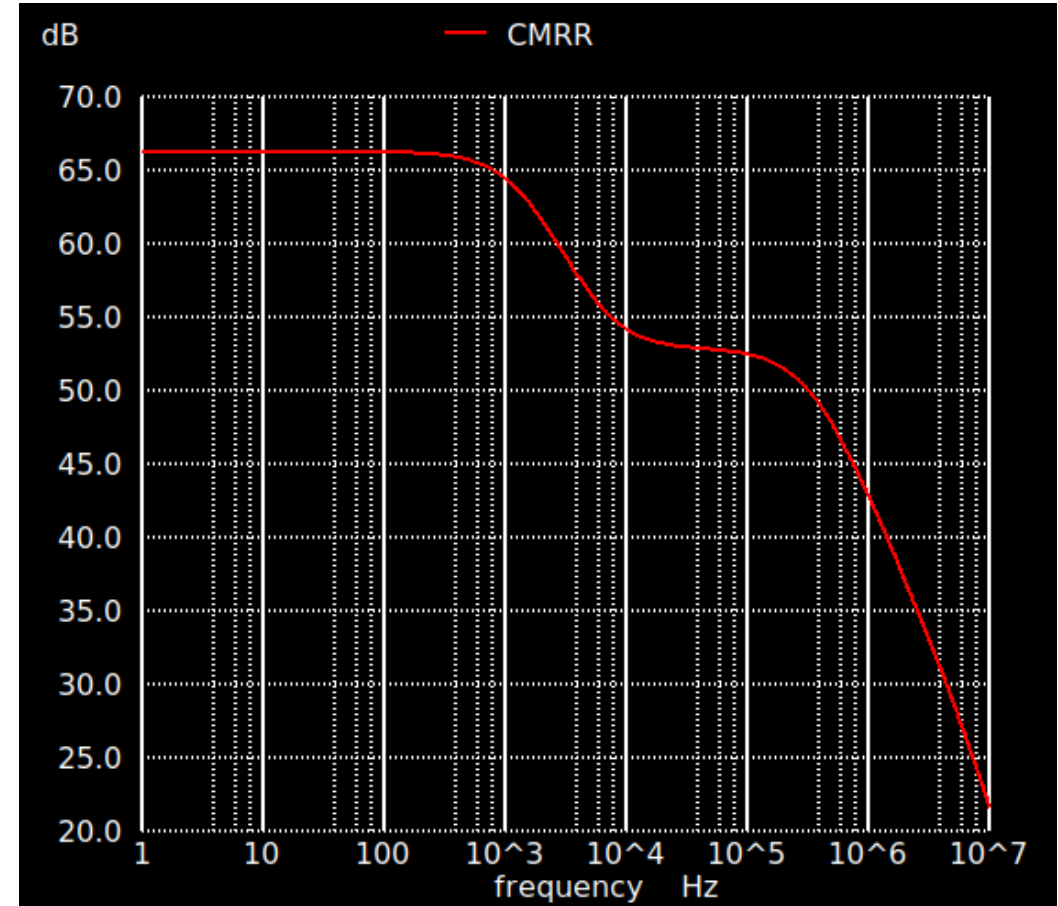
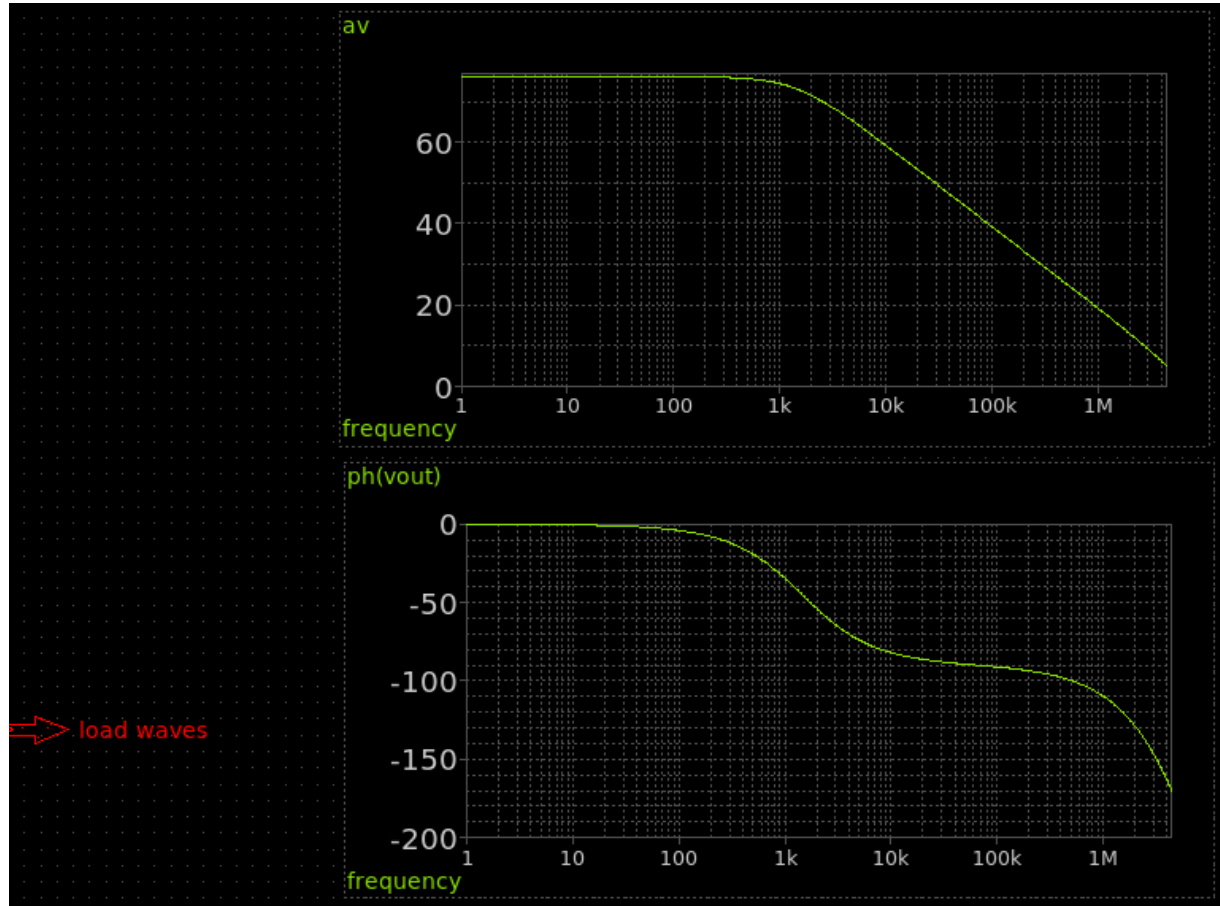
Select gm/id: 10.513728357868652 Set gm/gds: 0

Correspondi... 0.39

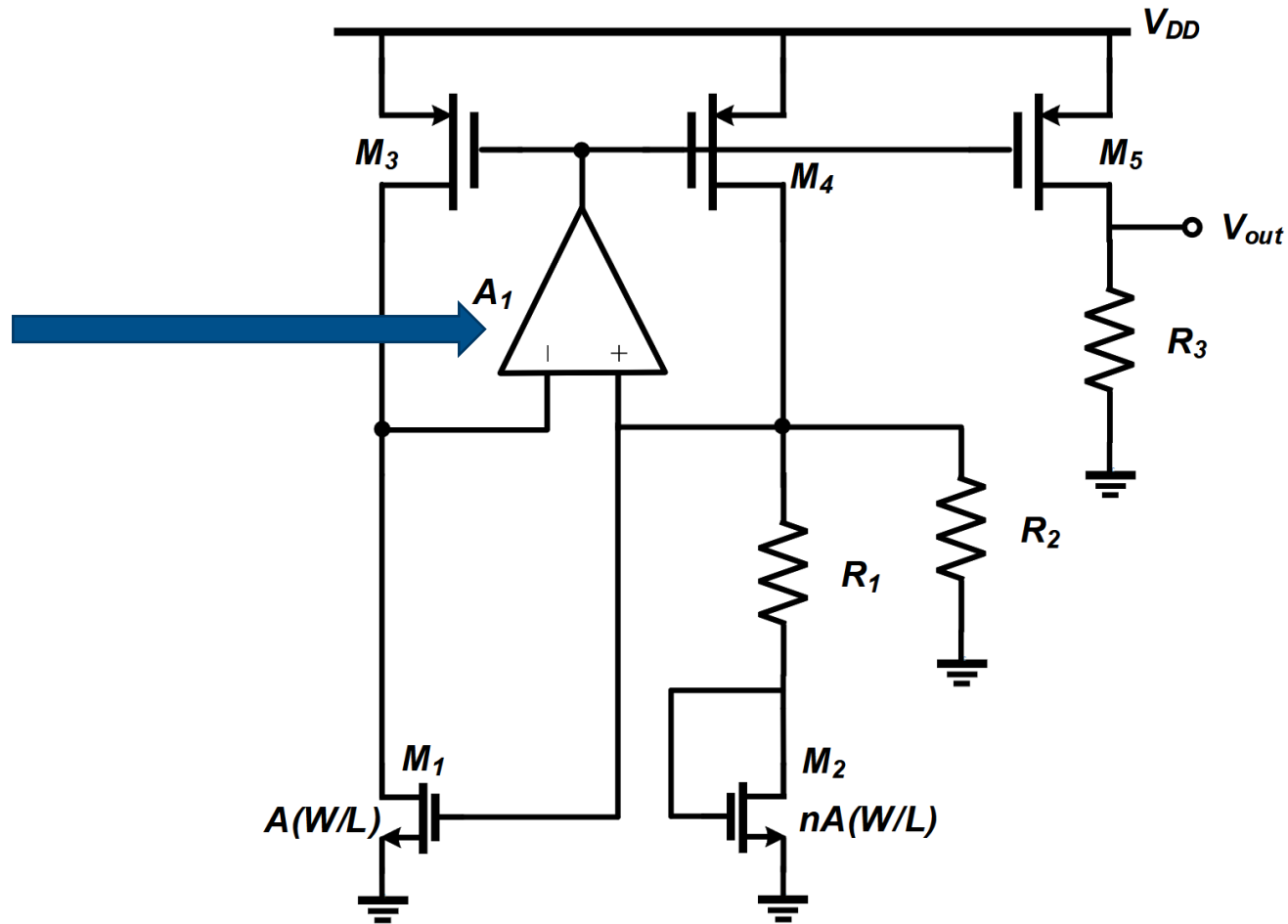
The corresponding gm/gds value for gm/id = 10.51 is: 42.06



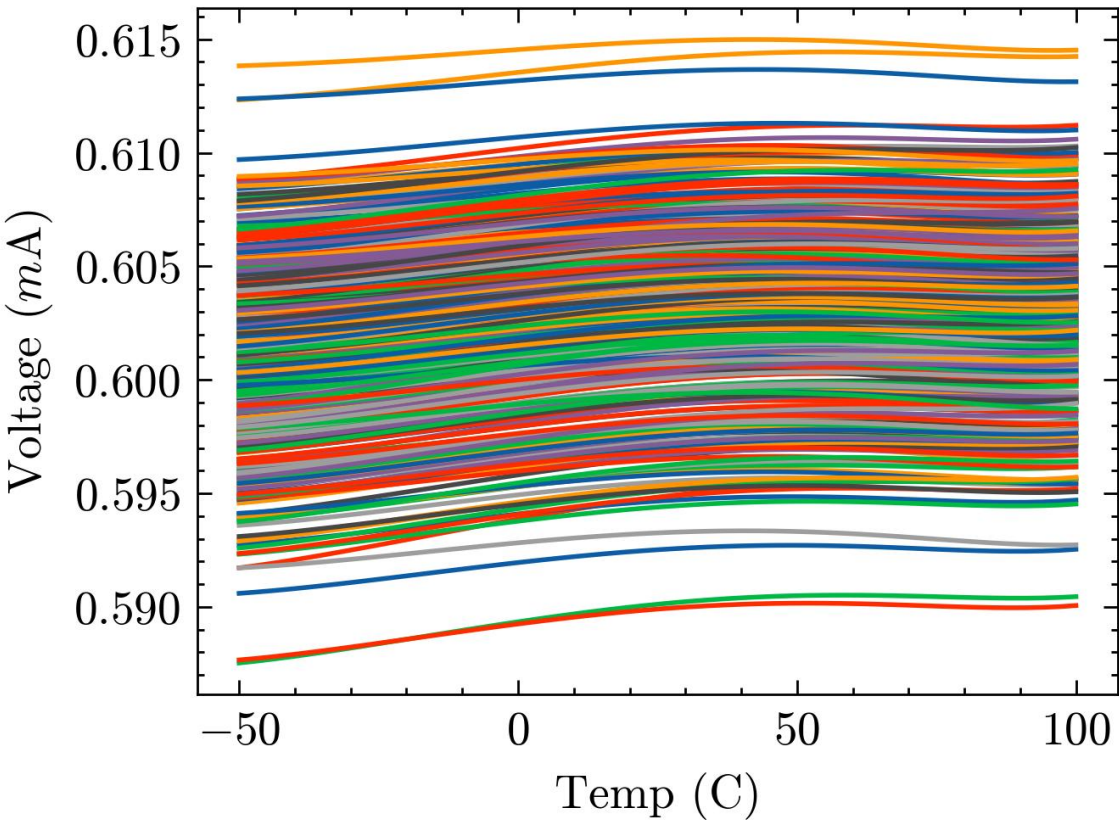
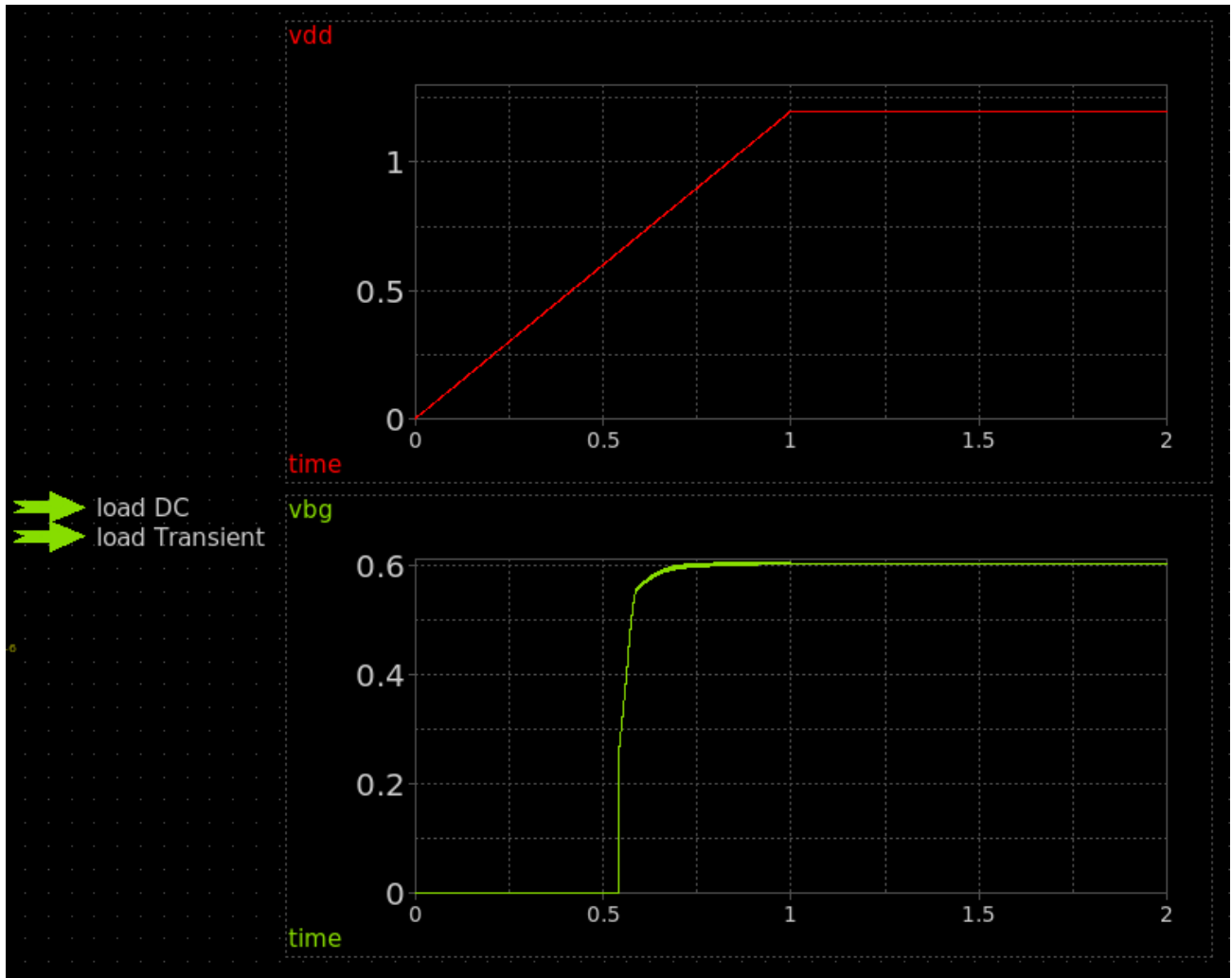
Testbenches



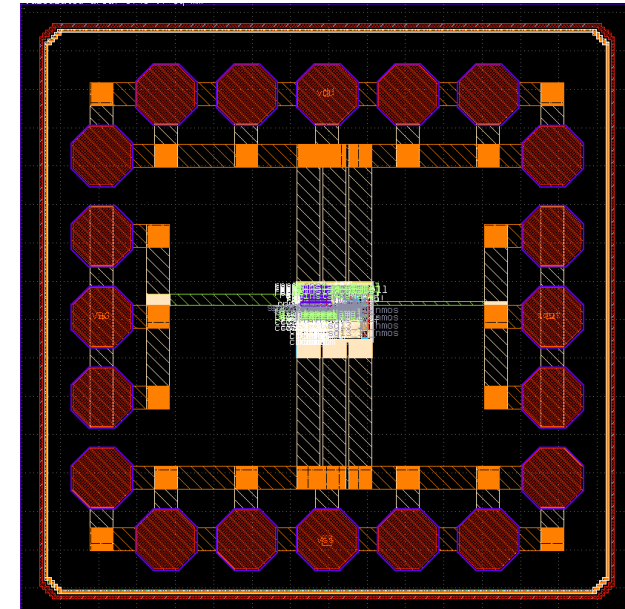
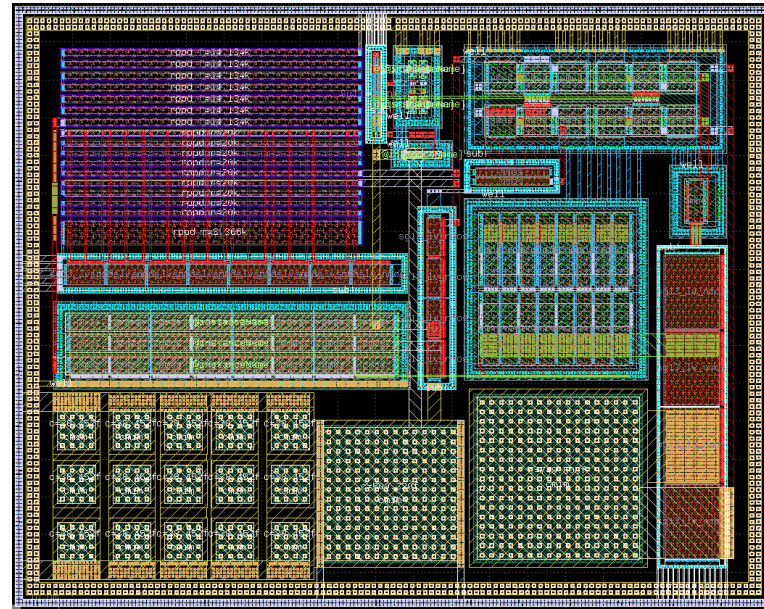
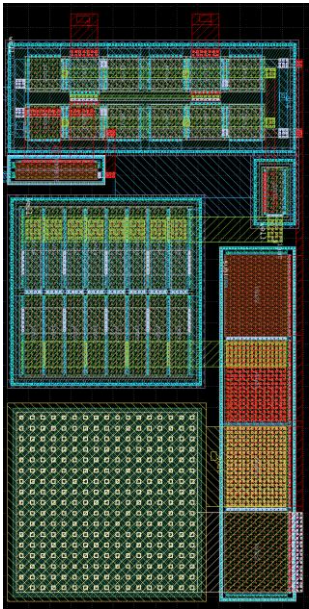
Bandgap Reference



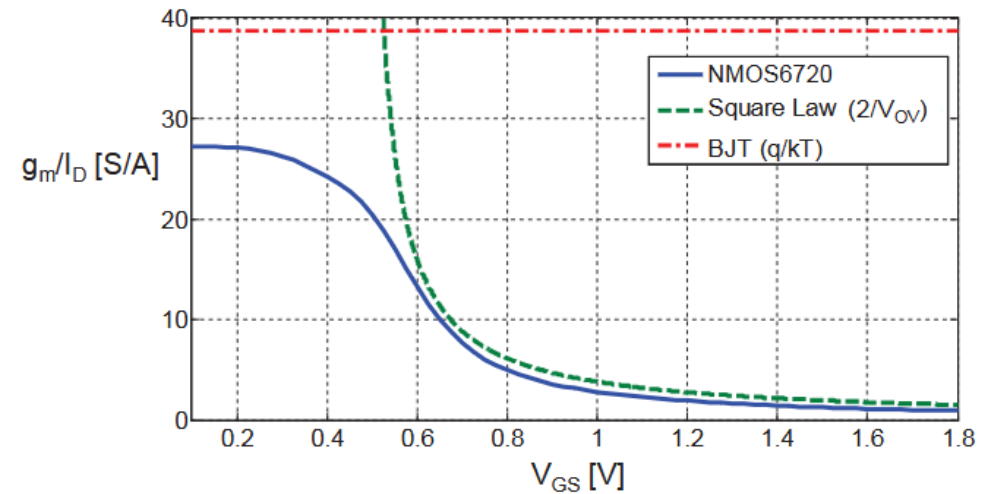
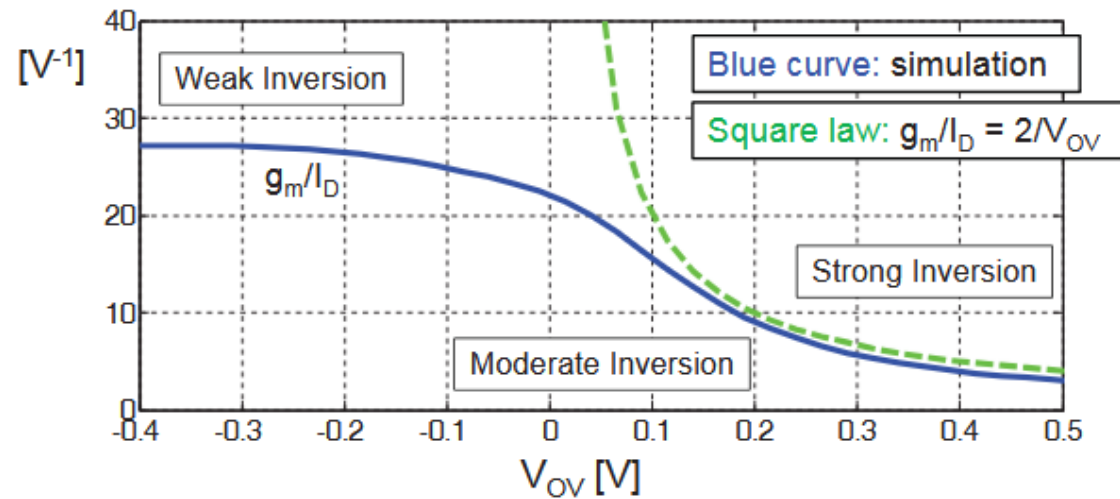
Evaluating Overall Performance



Bandgap Reference (Layout)



Small Insights In The Design Phase



R. Walker

ECE/CS 5720/6720 Fall 2017 – Chapter 5

3

Ross Walker ECE/CS 5720/6720 Fall 2017 University of Utah Partly adapted from Stanford's analog circuit design sequence

Small Insights In The Design Phase (Live Demo)



This screenshot shows the GitHub interface for the repository 'gmid' by user 'medwatt'. The repository is public and has 1 branch and 0 tags. The commit history shows a commit by 'medwatt' titled 'added automated 5T-OTA design' from 2 days ago, with 26 commits in total. The file list includes 'figures', 'src/mosplot', 'tests', '.gitignore', 'README.md', and 'setup.py'. Two red arrows point from the 'README.md' and 'setup.py' files in this repository to the corresponding files in the second repository shown.

This screenshot shows the GitHub interface for the repository 'update markdwon for gmid' by user 'PhillipRambo'. The repository contains a file list with folders 'inverter' and 'scripting', and files 'README.md', 'foundations.md', 'sg13_nmos_lv.py', and 'sg13_pmos_lv.py'. Two red arrows point from the 'README.md' and 'setup.py' files in the first repository to these files in this repository.

Original Repository: <https://github.com/medwatt/gmid>

Small Insights In The Design Phase



```
from mosplot.lookup_table_generator.simulators import NgspiceSimulator, HspiceSimulator
from mosplot.lookup_table_generator import LookupTableGenerator, TransistorSweep
# One of 'include_paths' or 'lib_mappings' must be specified.
# The rest are optional.

ngspice = NgspiceSimulator(
    # Provide path to simulator if simulator is not in system path.
    simulator_path="ngspice",

    # Default simulation temperature. Override if needed.
    temperature=27,

    # All parameters are saved by default. Override if needed.
    parameters_to_save=["id", "vth", "vdsat", "gm", "gds", "vgs"],

    # Files to include with '.LIB'.
    lib_mappings = [
        ("/home/pedersen/IHP-Open-PDK/ihp-sg13g2/libs.tech/ngspice/models/cornerMOSlv.lib", " mos_tt") # Put your own path to the corner lib
    ],

    # If the transistor is defined inside a subcircuit in
    # the library files, you must specify the symbol used (first entry)
    # and the hierarchical name (second entry). Override if needed.
    mos_spice_symbols = ("XM1", "n.xm1.ngs13_lv_nmos"),

    # Specify the width. For devices that do not take a width,
    # you can specify other parameters such as the number of fingers.
    # The keys are exactly those recognized by the model.
    device_parameters = {
        "w": 10e-6,
    }
)

nmos_sweep = TransistorSweep(
    mos_type="nmos",
    vgs=(0, 1.2, 0.01),
    vds=(0, 1.2, 0.01),
    vbs=(0, -1.2, -0.1),
    length = [130e-9, 260e-9, 390e-9, 520e-9, 650e-9, 780e-9, 910e-9, 1040e-9, 1170e-9, 1300e-9, 1430e-9, 1560e-9, 1690e-9, 1820e-9, 1950e-9, 2080e-9, 2210e-9, 2340e-9, 2470e-9, 2600e-9, 2730e-9,
2860e-9, 2990e-9, 3120e-9, 3250e-9, 3380e-9, 3510e-9, 3640e-9, 3770e-9, 3900e-9, 4030e-9, 4160e-9, 4290e-9, 4420e-9, 4550e-9, 4680e-9, 4810e-9, 4940e-9, 5070e-9, 5200e-9, 5330e-9, 5460e-9, 5590e-9,
5720e-9, 5850e-9, 5980e-9, 6110e-9, 6240e-9, 6370e-9, 6500e-9, 6630e-9, 6760e-9, 6890e-9, 7020e-9, 7150e-9, 7280e-9, 7410e-9, 7540e-9, 7670e-9, 7800e-9, 7930e-9, 8060e-9, 8190e-9, 8320e-9, 8450e-9,
8580e-9, 8710e-9, 8840e-9, 8970e-9, 9100e-9, 9230e-9, 9360e-9, 9490e-9, 9620e-9, 9750e-9, 9880e-9]
)
```

[IHP-AnalogAcademy/modules/module_0_foundations/sg13_nmos_lv.py](#)

[IHP-AnalogAcademy/modules/module_0_foundations/sg13_pmos_lv.py](#)

Small Insights In The Design Phase



- 0 Enter the virtual environment **enter_venv**
- 0 Run the python scripts to generate the lookup tables: **python3 sg13_nmos_lv.py** - also for PMOS
- 0 Navigate to *modules/module_1_bandgap_reference/part_1_OTA/gmid_example/scripting*
- 0 Open the Jupyter lab file and try to visualize the results as seen in the next slide

jupyter lab OTA_low_gain.ipynb

Small Insights In The Design Phase (Live Demo)



```
[101]: import matplotlib.pyplot as plt
import numpy as np
from mosplot.plot import load_lookup_table, Mosfet, Expression
import ipywidgets as widgets
from ipywidgets import interactive
from ipywidgets import interactive_output, HBox, VBox
import matplotlib.ticker as ticker

[102]: lookup_table_nmos = load_lookup_table('../sg13_nmos_lv_LUT.npz')
lookup_table_pmos = load_lookup_table('../sg13_pmos_lv_LUT.npz')

[103]: print(lookup_table.keys())

dict_keys(['sg13_lv_nmos ', 'sg13_lv_pmos', 'description', 'simulator', 'parameter_names', 'device_parameters'])

[108]: nmos = Mosfet(lookup_table=lookup_table_nmos, mos="sg13_lv_nmos ", vbs=0.0, vds=0.6)
pmos = Mosfet(lookup_table=lookup_table_pmos, mos="sg13_lv_pmos", vbs=0.0, vds=-0.6, vgs=(-1.2, -0.15))

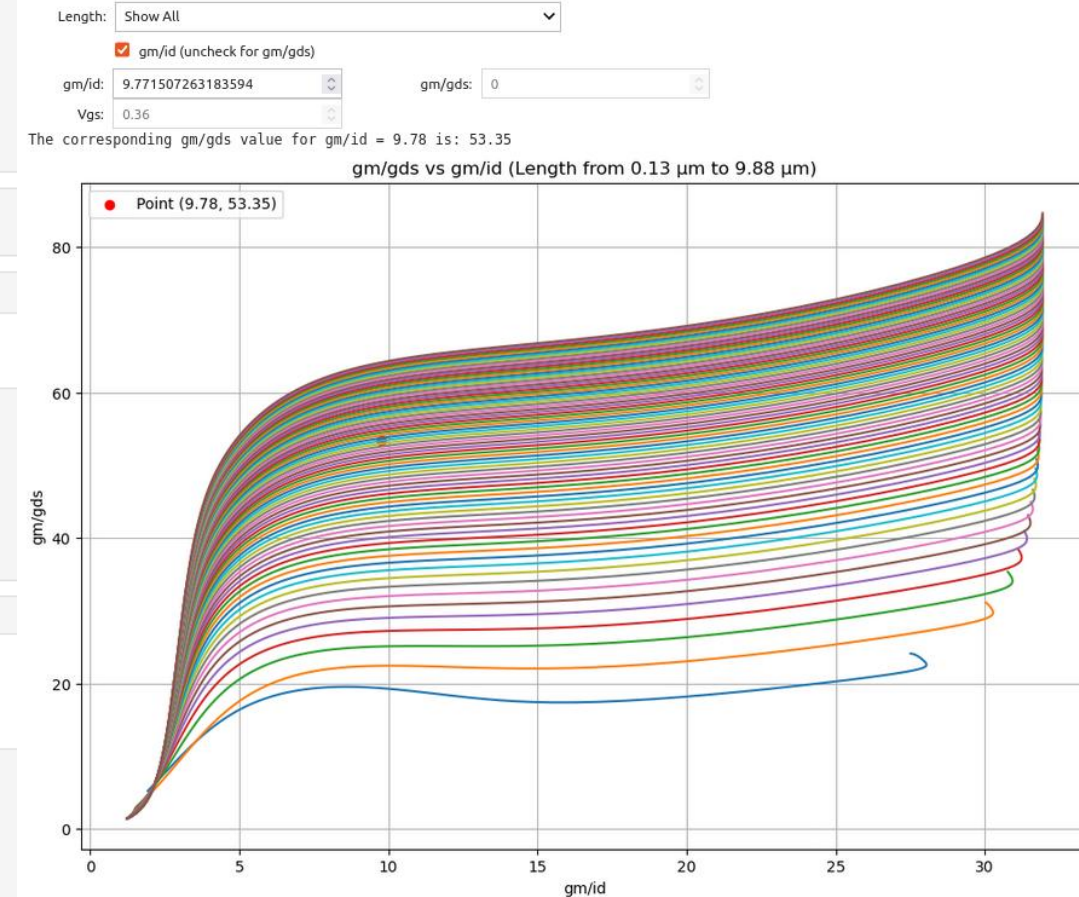
rows_0, cols_0 = np.shape(nmos.extracted_table['gm']) # just for getting the shape of the data
rows_1, cols_1 = np.shape(pmos.extracted_table['gm']) # just for getting the shape of the data
reshaped_lengths_nmos = np.tile(nmos.length[:, np.newaxis], (1, cols_0))
reshaped_lengths_pmos = np.tile(pmos.length[:, np.newaxis], (1, cols_1))

def plot_data_vs_data(x_values, y_values, z_values, length, x_axis_name, y_axis_name='y', y_multiplier=1, log=False): ...

NMOS GMID

[110]: width_values = nmos.width
id_values = nmos.extracted_table['id']
gm_values = nmos.extracted_table['gm']
gds_values = nmos.extracted_table['gds']
vgs_values = nmos.extracted_table['vgs']

plot_data_vs_data(gm_values/id_values, gm_values/gds_values, vgs_values, reshaped_lengths_nmos, 'gm/id', 'gm/gds')
```





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Part 1

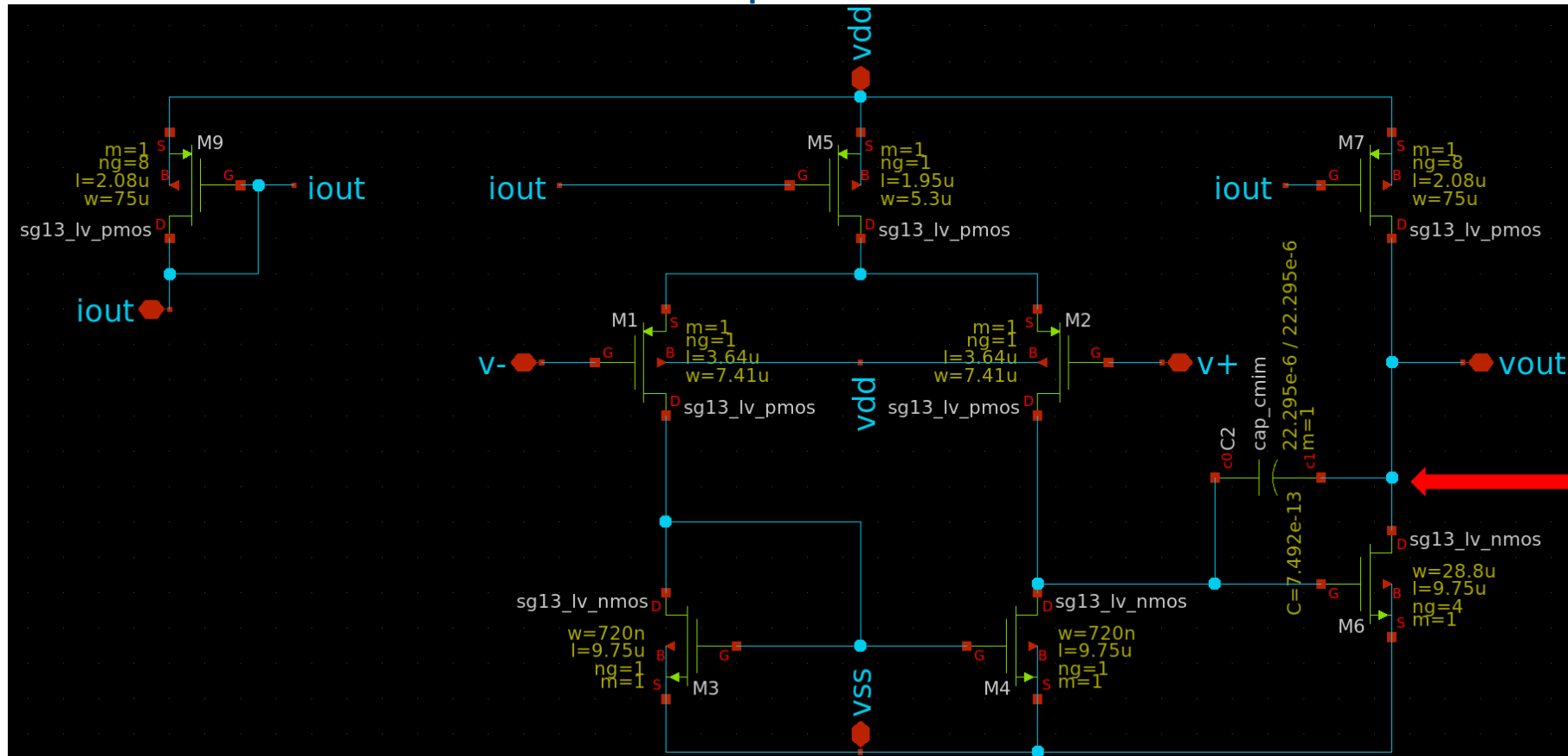
Miller Compensated OTA-Design

Reference: /modules/module_1_bandgap_reference/part_1_OTA

Schematic & Symbol



- 0 Create a folder for this module and organize yourself within
- 0 Create a Schematic for the OTA seen in the picture



CMIM

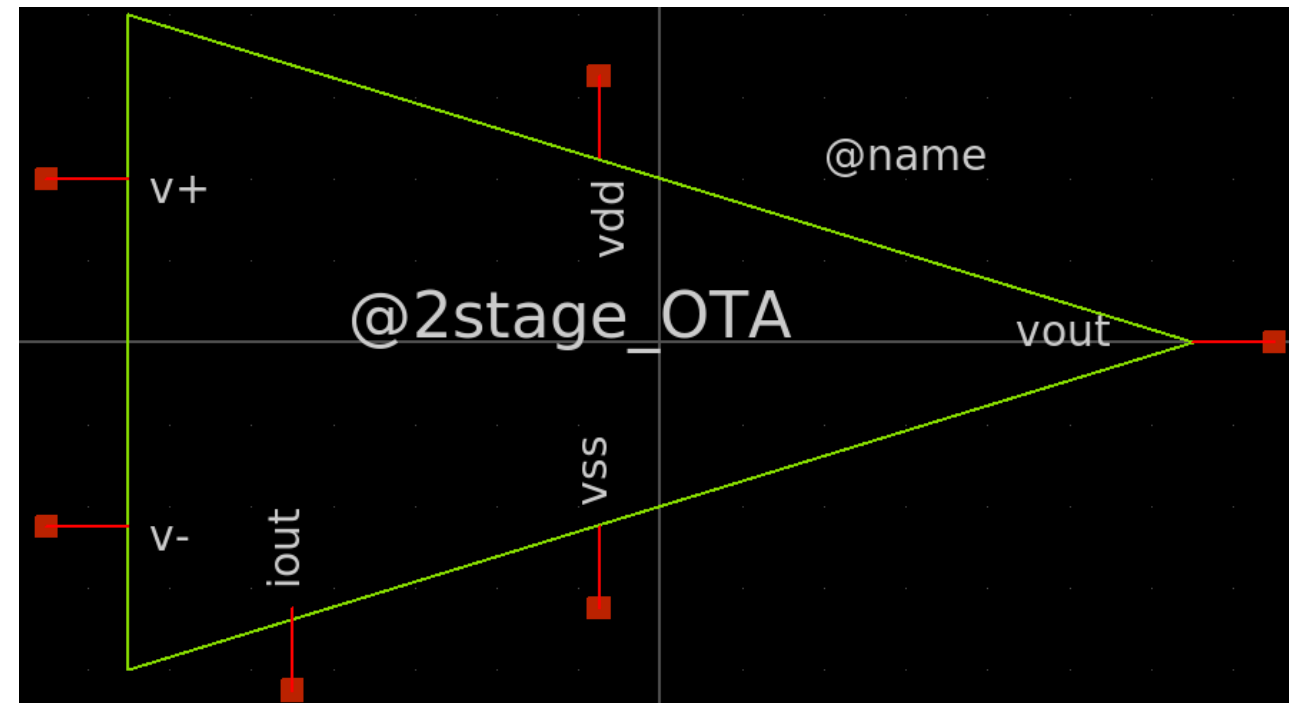
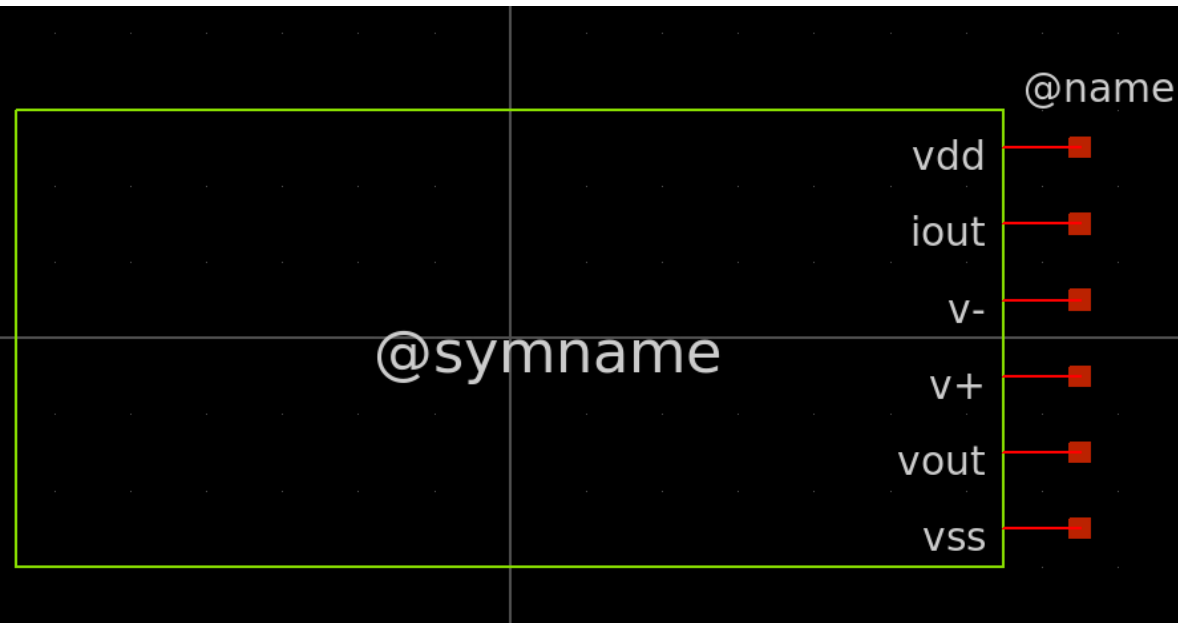
Specs cmim:

$W = L = 22.295e-6$

Schematic & Symbol



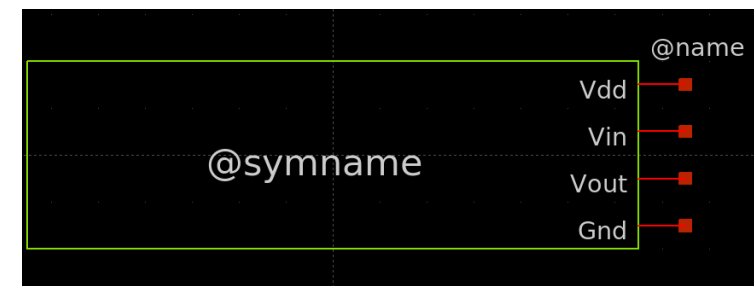
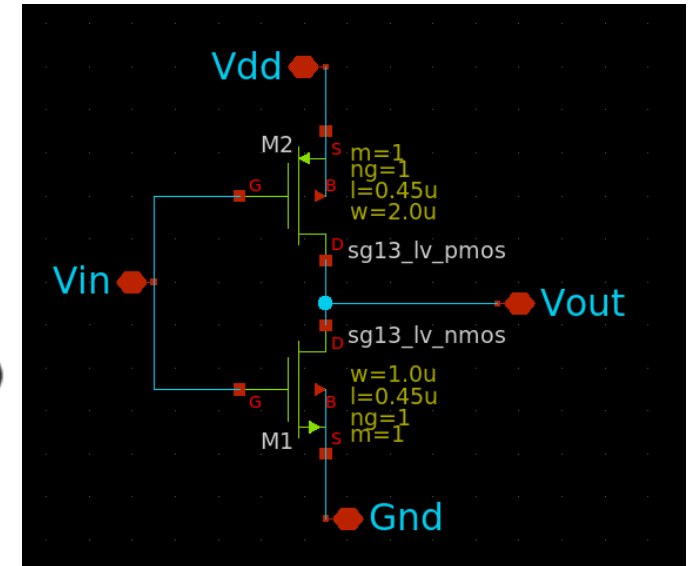
- Create the OTA symbol in your preferred style
- The next slide provides a reminder on how to create a symbol



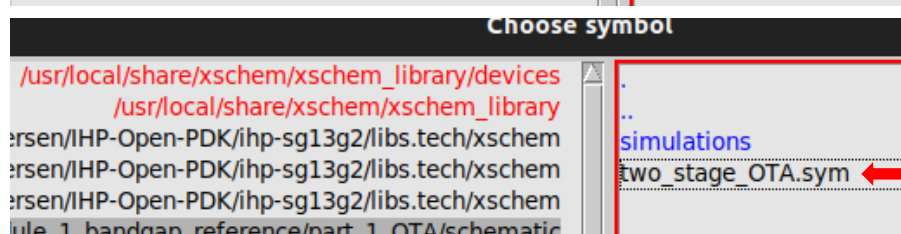
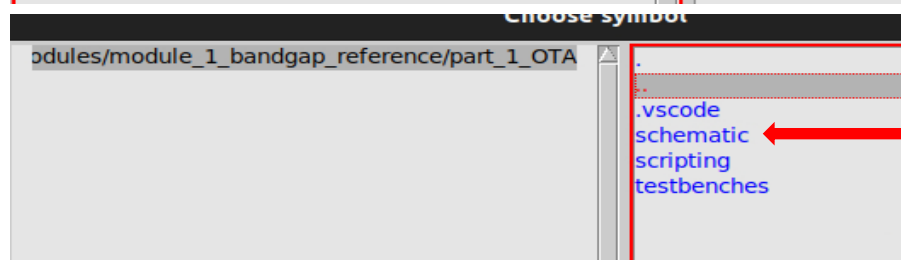
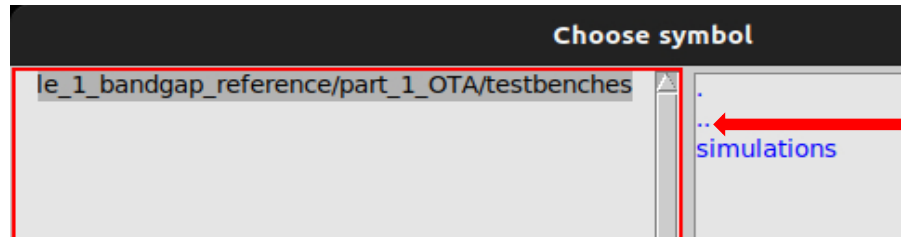
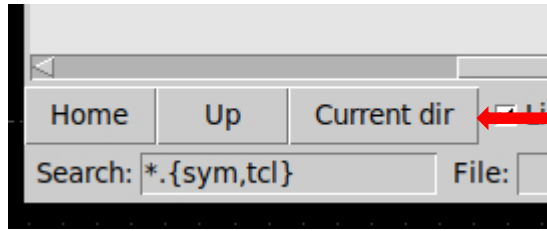
Creating a Symbol (from yesterday)



- 0 For Creating the Symbol press **A**
- 0 This creates a symbol file in the directory of the schematic, navigate to this file and open it by writing (**xschem inverter.sym**)
- 0 Now you can drag the already existing lines to create the boundaries or use the tools seen in the toolbar to define custom shapes

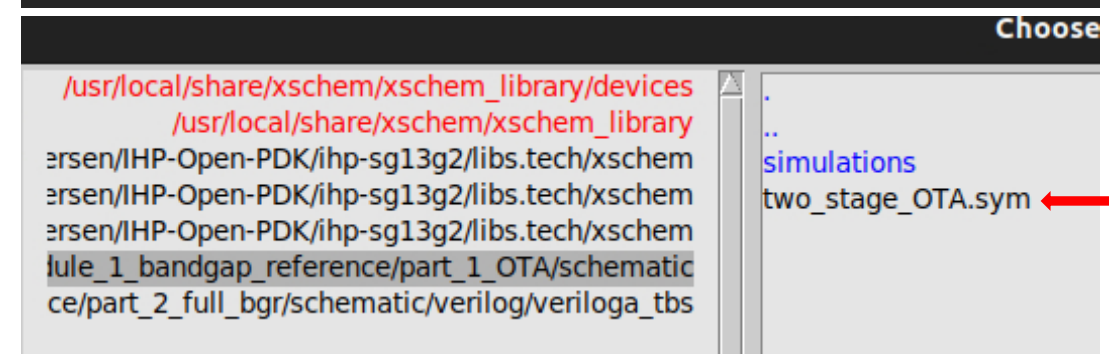


Accessing the Symbol (Two Options)

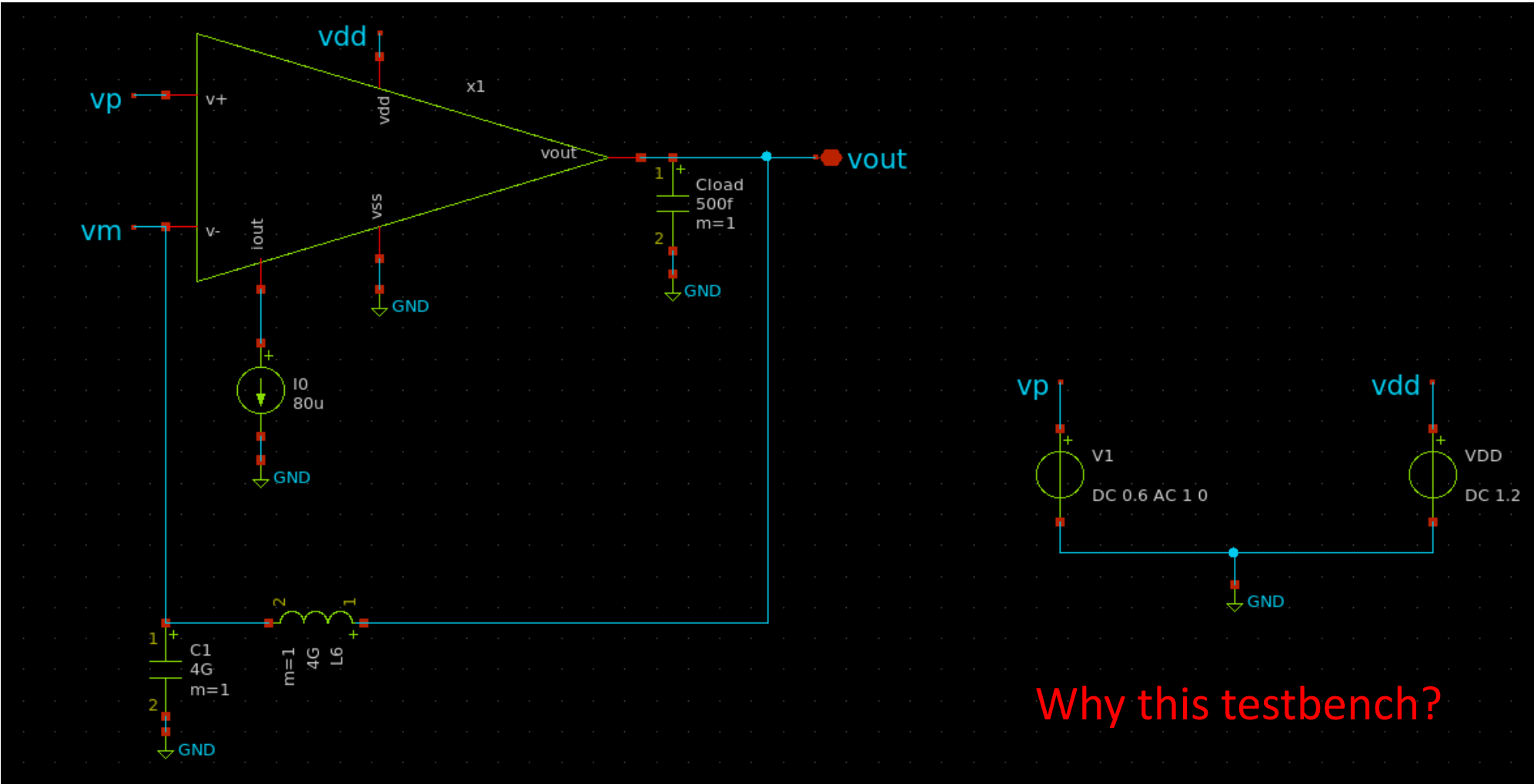


```
→ testbenches git:(main) X ls  
ota_testbench_mc_mis.sch ota_testbench.sch simulations xschemrc
```

```
Open ▾ ~/projects/IHP-AnalogAcademy/modules/module_1_bandgap_reference/part_1_OTAs/testbenches Save ≡  
1 # xschemrc - Custom configuration file for xschem  
2 # This file sources another xschemrc file from a known location  
3  
4 # Source the base configuration from a known location  
5 source $::env(PDK_ROOT)/$::env(PDK)/libs.tech/xschem/xschemrc  
6  
7 # (Optional) Add any custom overrides or extensions below  
8 # set xschem_library_path /home/user/my_libs  
9 # set xschem_gui_font "Monospace 10"  
10  
11 ##### include skywater libraries. Here I use [pwd]. This works if I start xschem from here.  
12 #####only if you dont have this setup already ###  
13 #####append XSCHM_LIBRARY_PATH :[file dirname [info script]]  
14  
15  
16 ##### Add custom libraries (directories with .lib files)  
17 append XSCHM_LIBRARY_PATH :$PDK_ROOT/ihp-sg13g2/libs.tech/xschem  
18 append XSCHM_LIBRARY_PATH :../schematic/ ←  
19 append XSCHM_LIBRARY_PATH :../../part_2_full_bgr/schematic/verilog/veriloga_tbs/
```



OTA Testbench



Why this testbench?

OTA Testbench: Finish The Testbench



Model Definition

Path: /usr/local/share/xschem/xschem_library/devices

Symbol devices/code_shown.sym

☐ No change properties ☐ Preserve unchanged props ☐ Copy cell

name=MODEL only_toplevel=true
format="tclevel(@value)"
value="
.lib \$::SG13G2_MODELS/cornerCAP.lib cap_typ
.lib cornerMOSlv.lib mos_tt
"

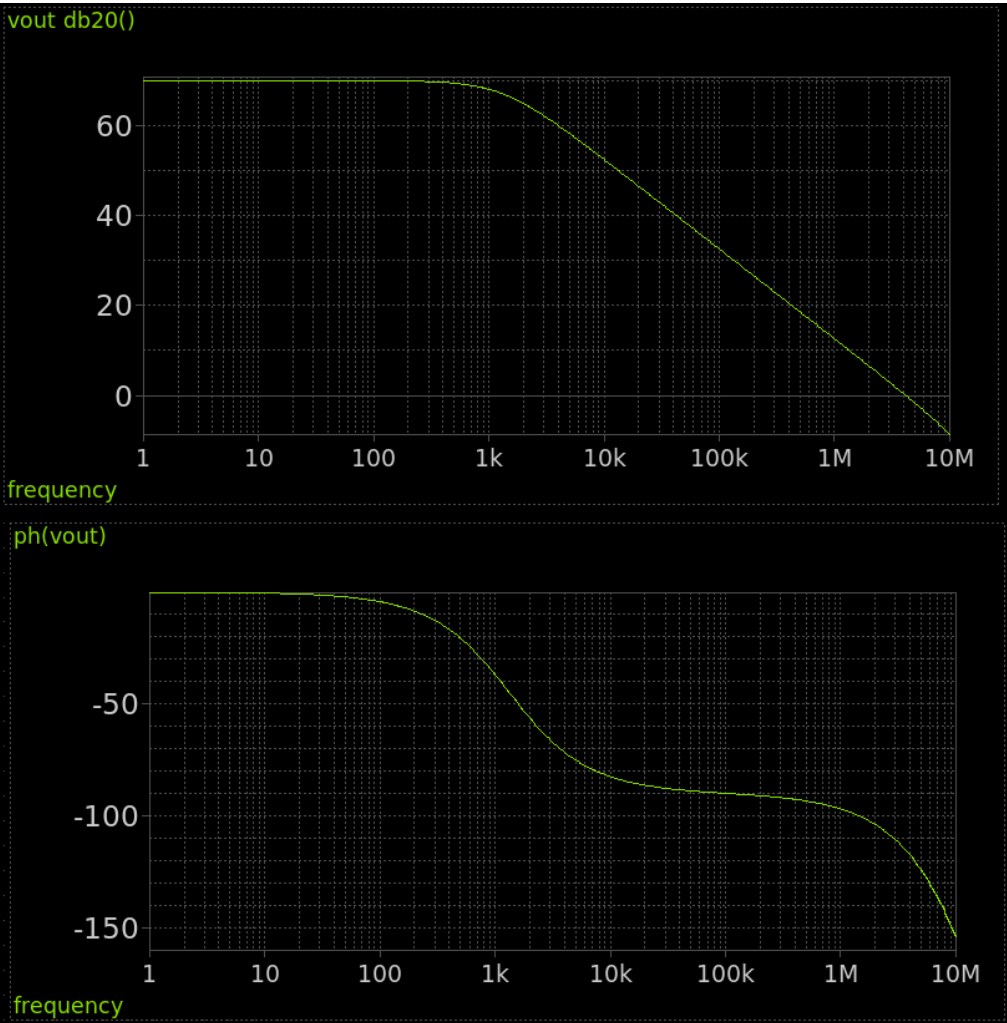
Simulation Code

```
name=NGSPICE only_toplevel=false
value="
.control
op
save all
write tb_OTA_op.raw
.endc

.control
op
ac dec 100 1 10e6
save all
let Av = db(v(vout))
write output_file.raw
.endc
"
```

DC Sim

AC Sim



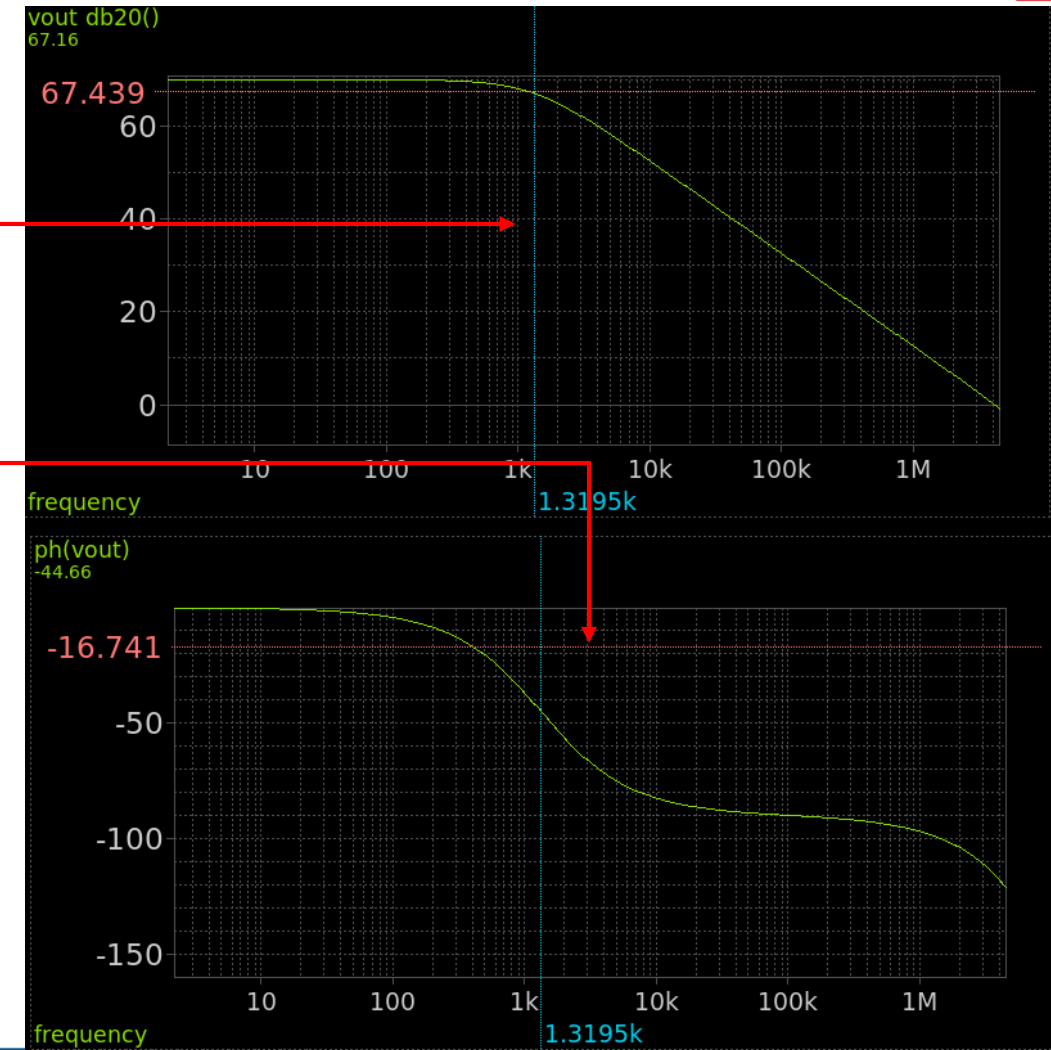
OTA Testbench: Finish The Testbench



Shortcuts for measuring in plot:

Vertical line: A

Horizontal Line: Shift + A



OTA Testbench: Op Annotation



Waves

External viewer

Clear

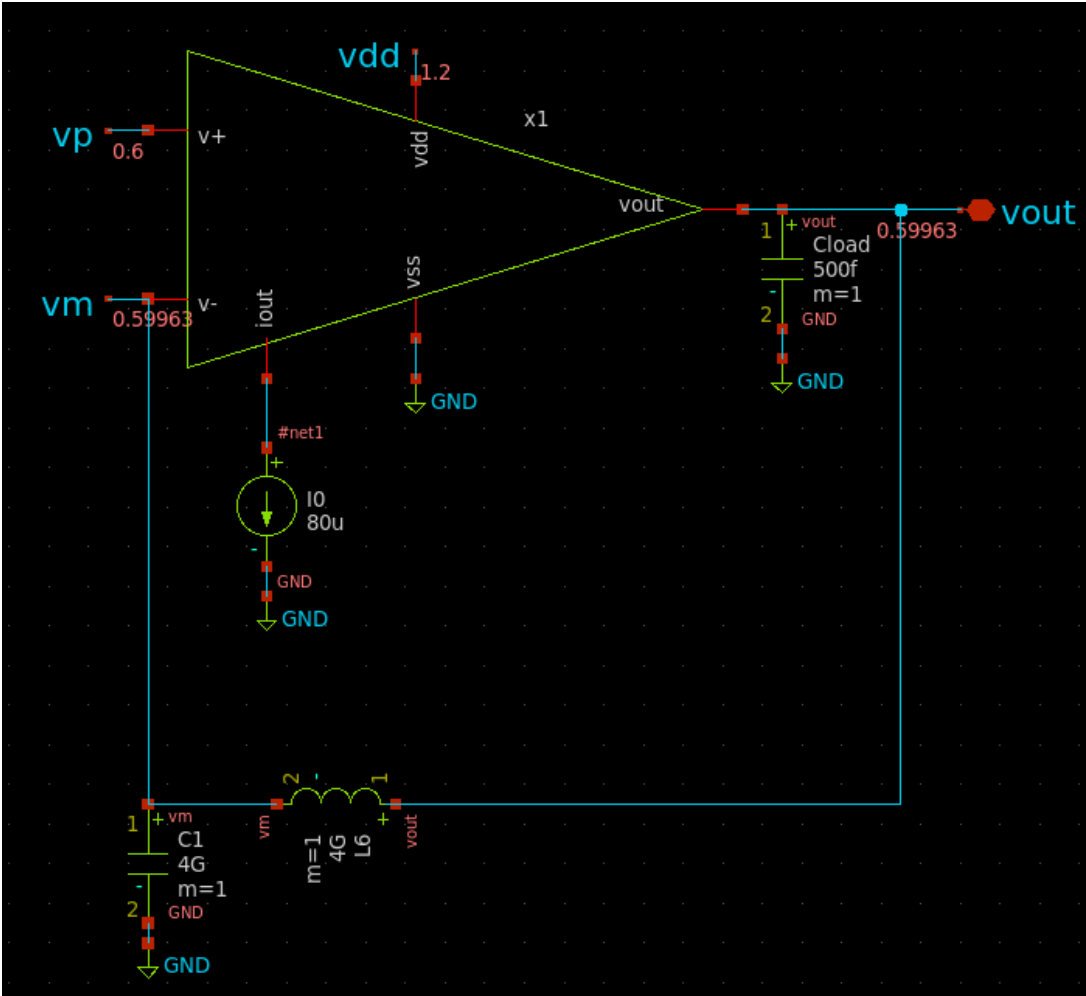
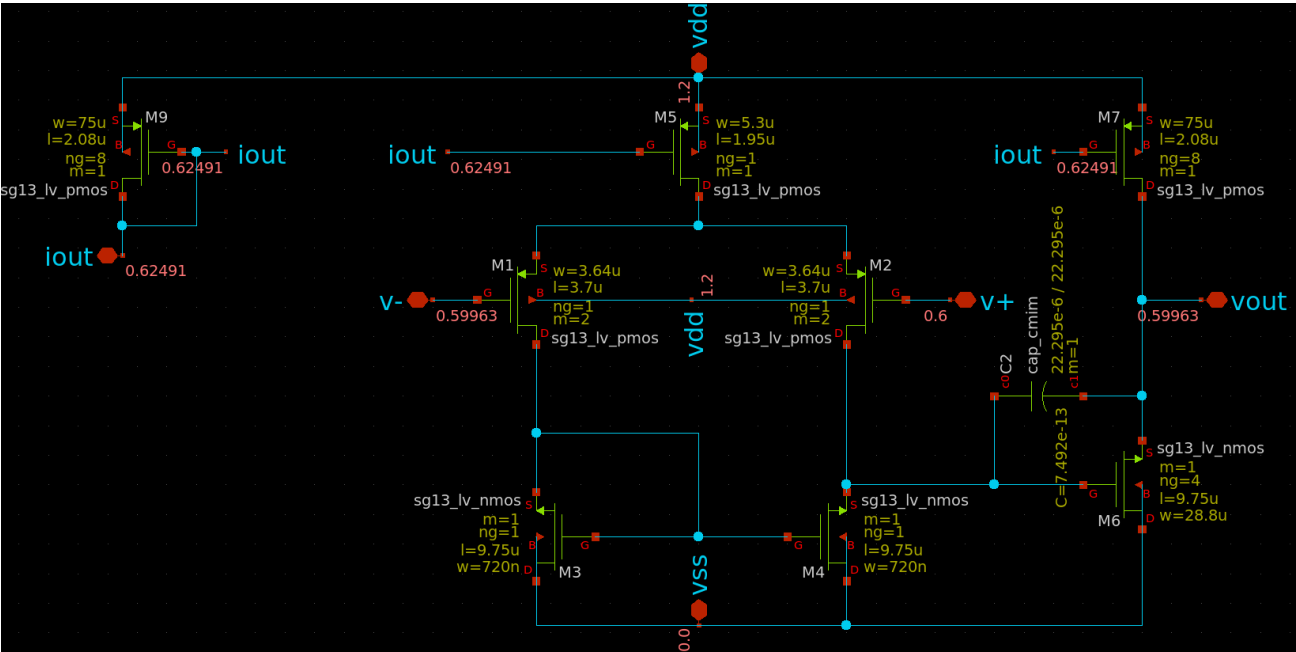
Op Annotate

Directory: /home/pedersen/proje

- ota_testbench_mc_mis.raw
- output_file.raw
- tb_OTA_op.raw

File name: tb_OTA_op.raw

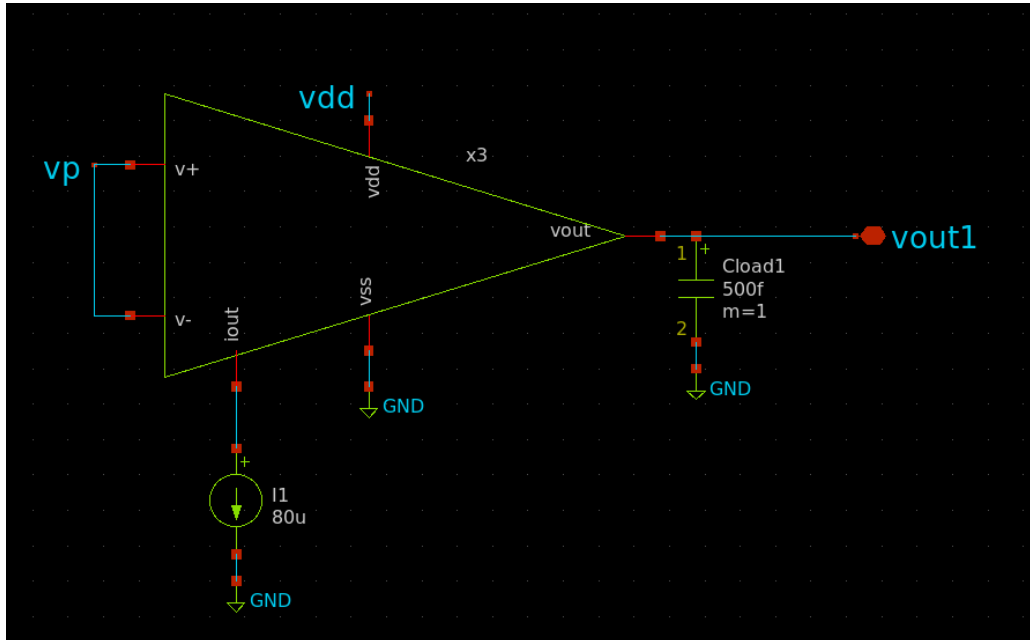
Files of type: Raw Files (*.raw)



OTA Testbench: CMRR & PSRR

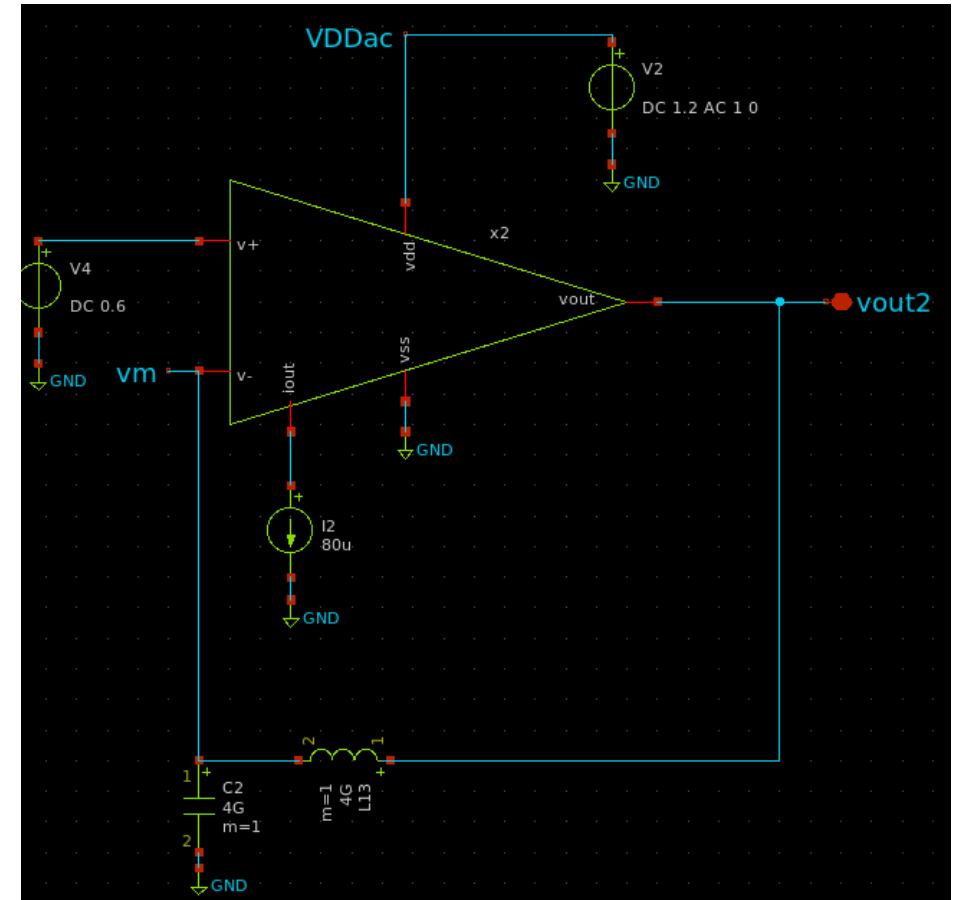


HINTS



$$CMRR_{dB} = 20 \log_{10} \left(\frac{A_{diff}}{A_{cm}} \right)$$

$$PSRR_{dB} = 20 \log_{10} \left(\frac{\Delta V_{out}}{\Delta V_{DD(AC)}} \right)$$



OTA Testbench: CMRR & PSRR



```
.control
op
ac dec 100 1 10e6
save all

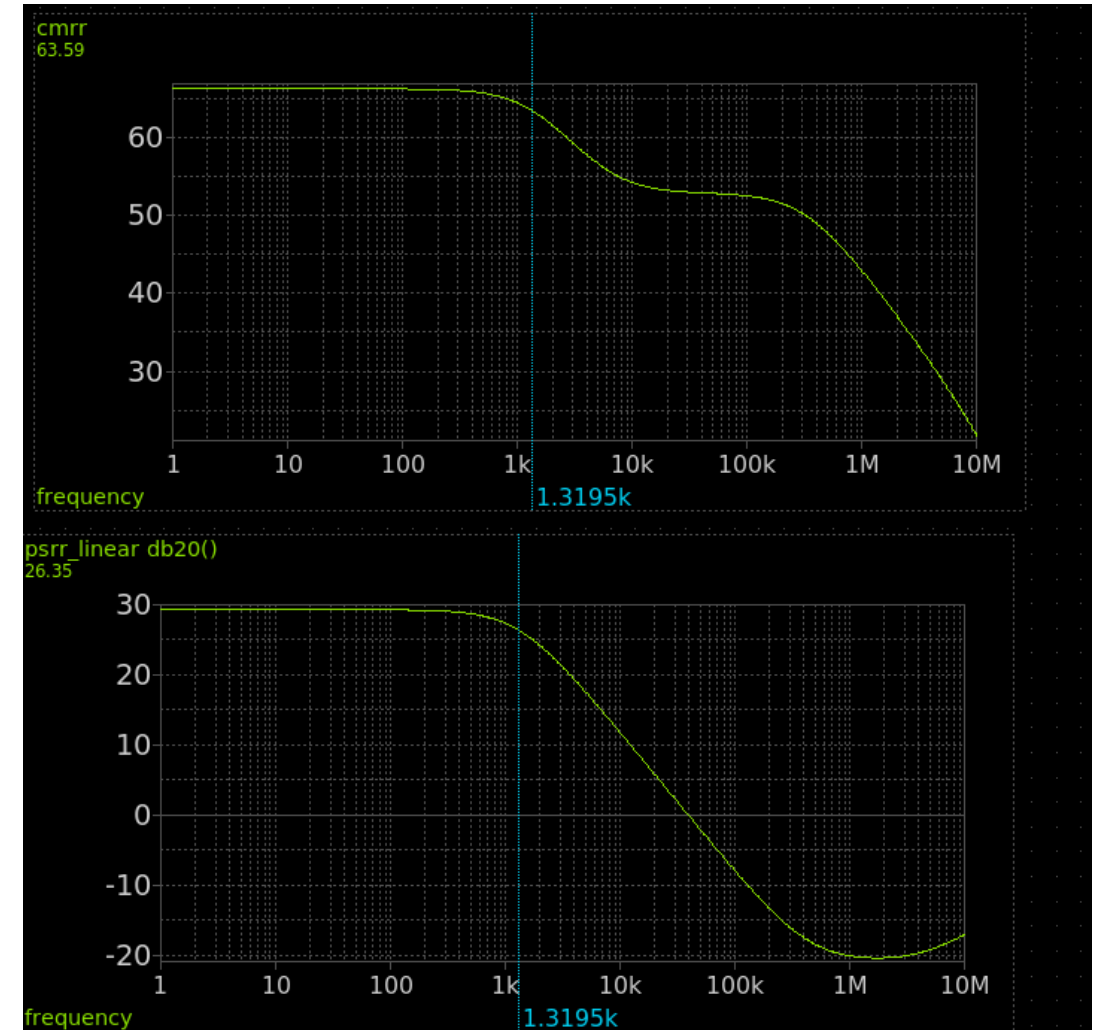
let Av = db(v(vout))

let PSRR_linear = v(vout2)/v(VDDac)

let CMRR = db((v(vout)/v(vp))/(v(vout1)/v(vp)))

let phase = 180*cph(vout)/pi

write output_file.raw
.endc
```





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Catching Up / Lunch Time !

A decorative graphic on the left side of the slide consisting of various colored lines (red, blue, grey) and small circles, resembling a circuit board layout or signal traces.

Next session:
Bandgap Reference Design



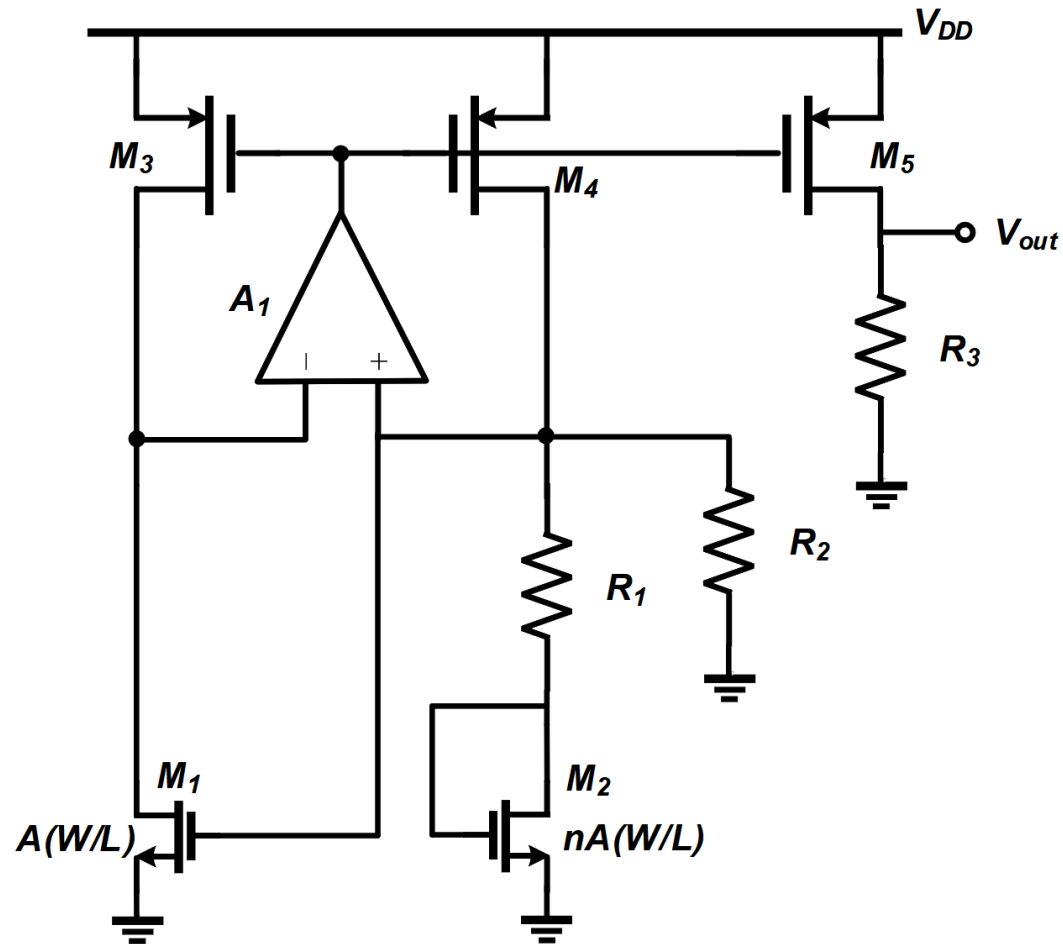
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Part 2

Bandgap Reference Design

Reference: /modules/module_1_bandgap_reference/part_2_full_bgr

Bandgap Reference

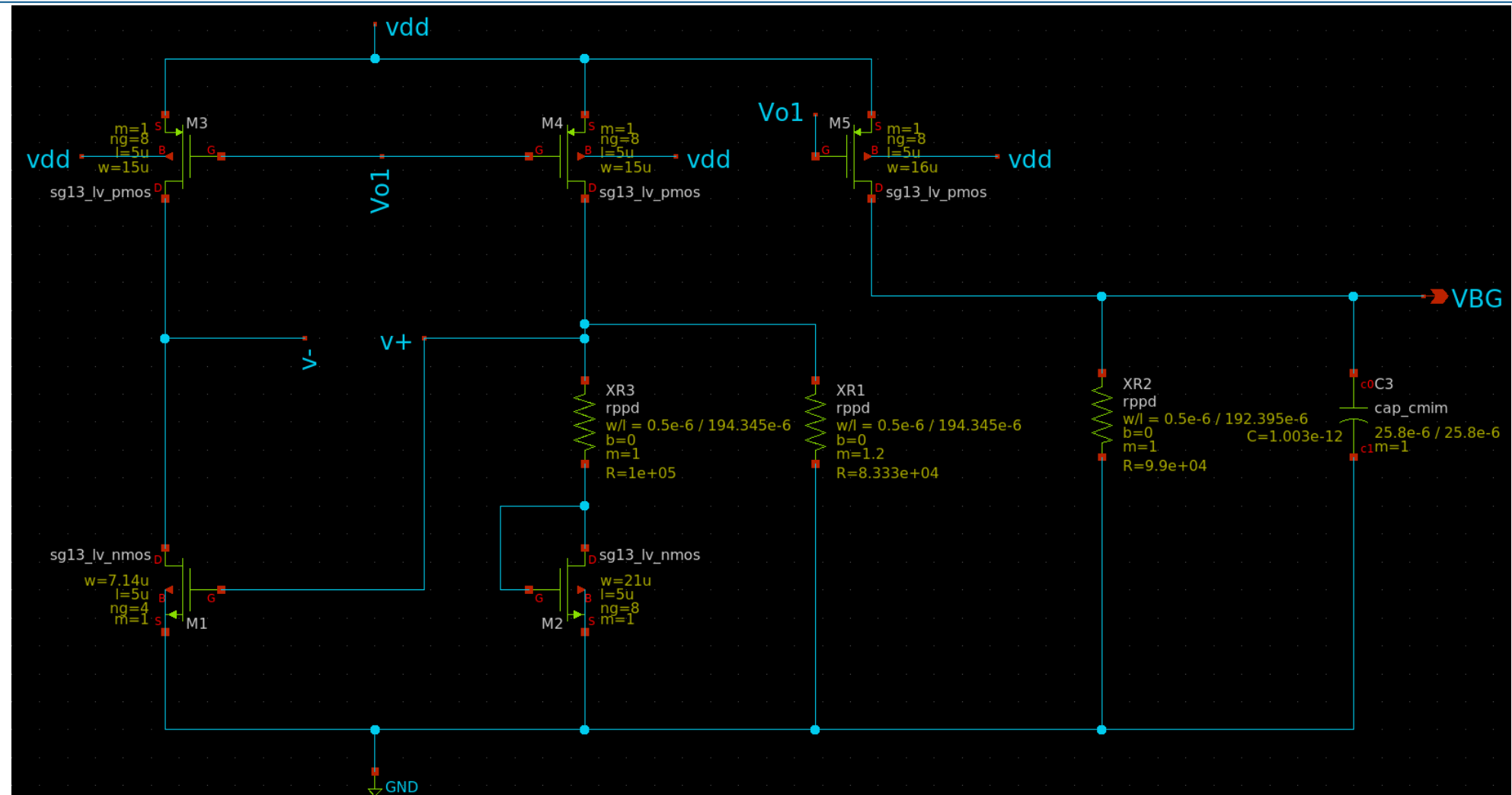


Bandgap Reference

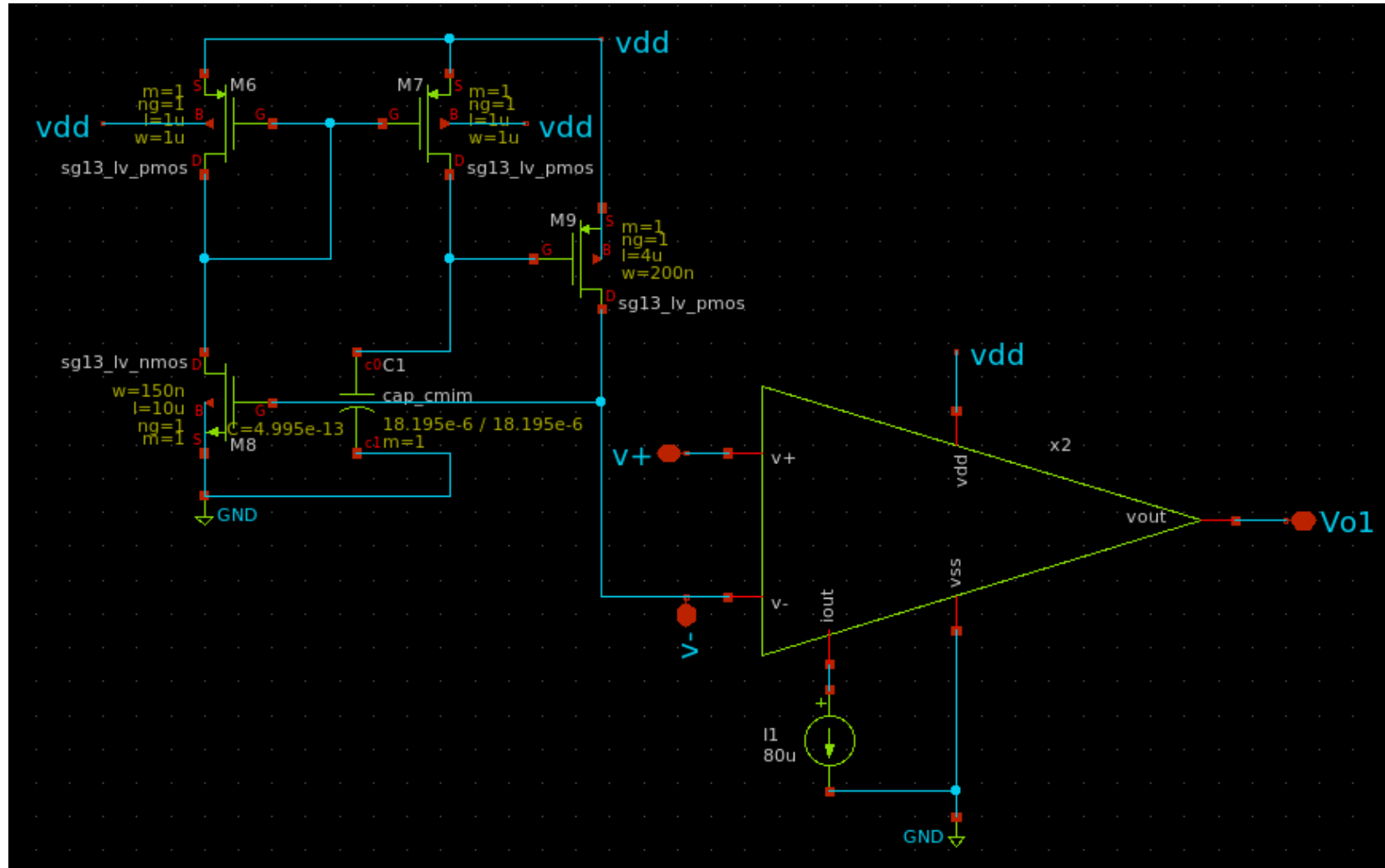


Resistor Type	Material	Sheet Resistance	Temperature Coefficient (TC)	Applications
Rsil	Salicided n-doped polysilicon	7 $\Omega/\square$	3100 ppm/K	Low-resistance paths, non-precision use
➔ Rppd	Unsalicided p-doped polysilicon	260 $\Omega/\square$	170 ppm/K	Precision and temperature-sensitive designs
Rhigh	High-resistance structures	Very high	Variable	High-impedance paths

Bandgap Reference Core Schematic



Start-up Circuit Schematic




```
name=MODEL only_toplevel=true  
format="tclevel( @value )" value="  
.lib $::SG13G2_MODELS/cornerCAP.lib cap_typ  
.lib $::SG13G2_MODELS/cornerRES.lib res_typ  
.lib cornerMOSlv.lib mos_tt "
```

DC Simulation

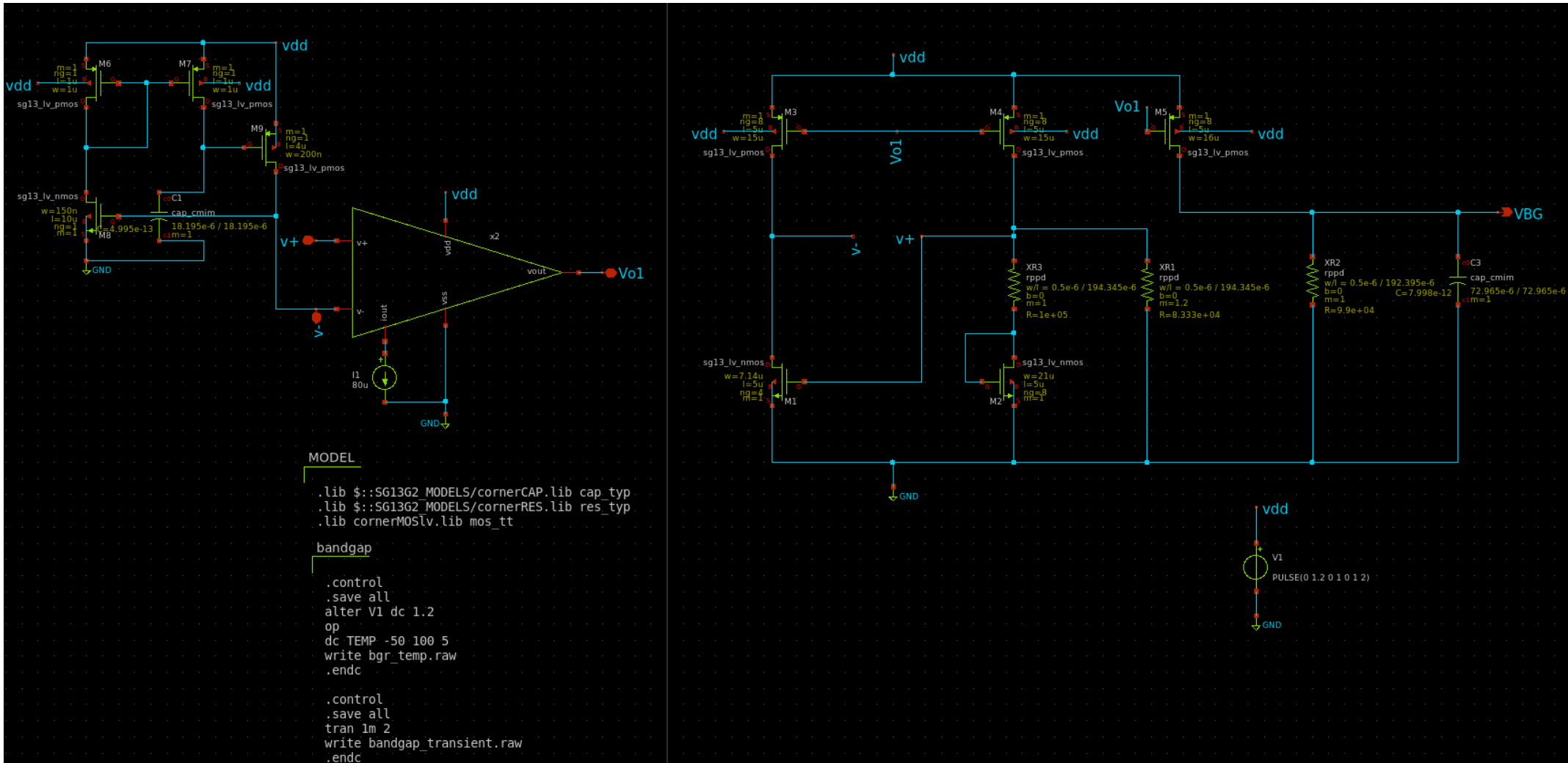
```
.control  
.save all  
alter V1 dc 1.2  
op  
dc TEMP -50 100 5  
write bgr_temp.raw .endc
```

Transient Simulation

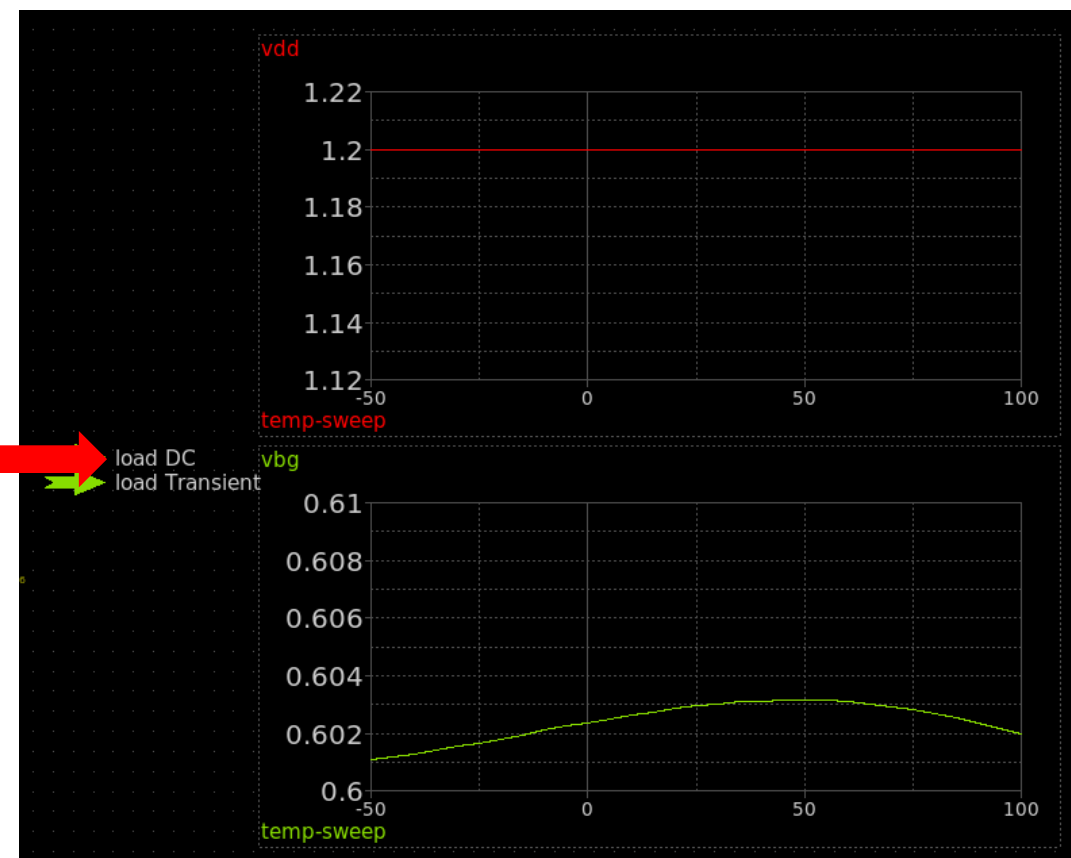
```
.control .save all  
tran 1m 2  
write bandgap_transient.raw .endc
```

VDD → "PULSE(0 1.2 0 1 0 1 2)"

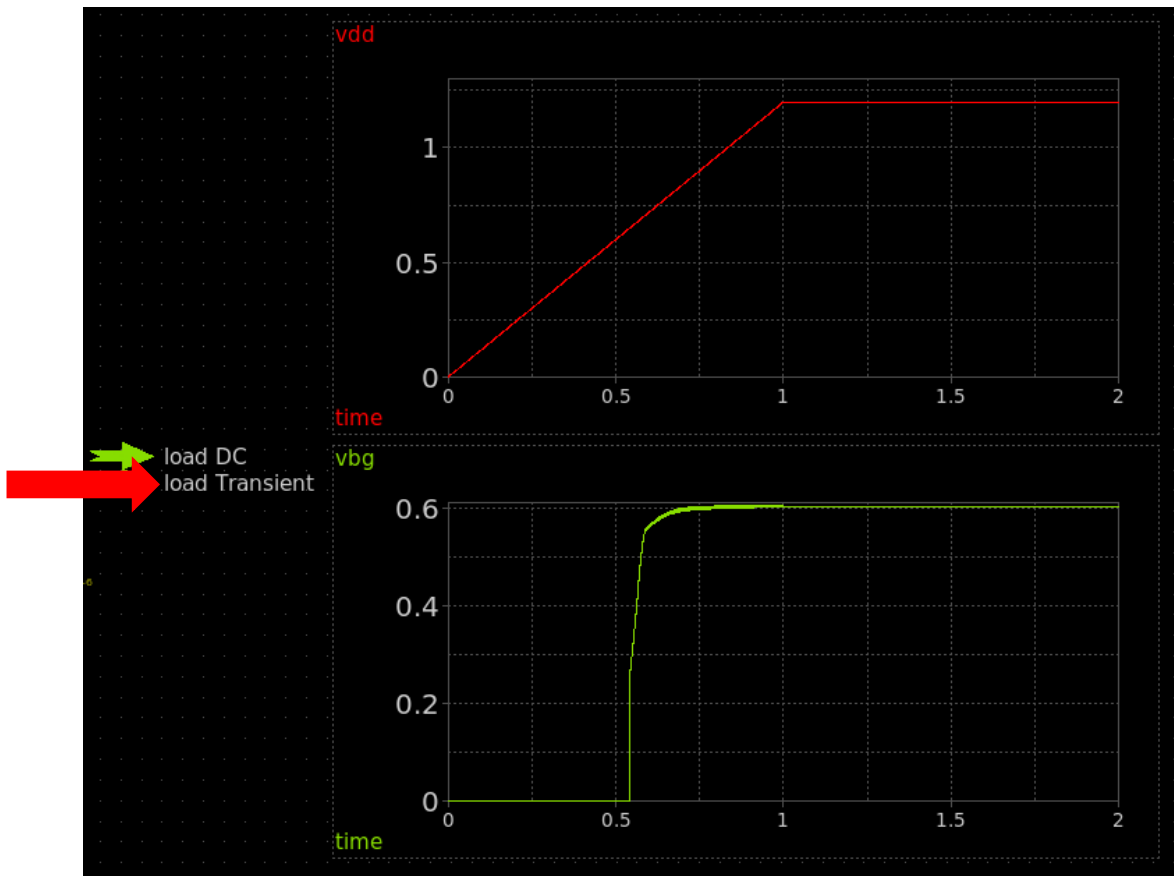
Simulation



bgr_temp.raw



bandgap_transient.raw



Mismatch in IC Manufacturing

- 0 Caused by **process variations**: geometry, doping, threshold voltage etc.
- 0 Critical in **symmetry-based circuits** (e.g., differential pairs, bandgap references).
- 0 Leads to **imbalanced currents**, voltage offsets, and **reduced accuracy**.

Monte Carlo Simulations: Why They Matter

- 0 Simulate **random device-level variations**.
- 0 Evaluate **circuit robustness** under realistic conditions.
- 0 **1 σ** (~68%), **2 σ** (~95%), **3 σ** (~99.7%) coverage of variation.

Mismatch Models and Code



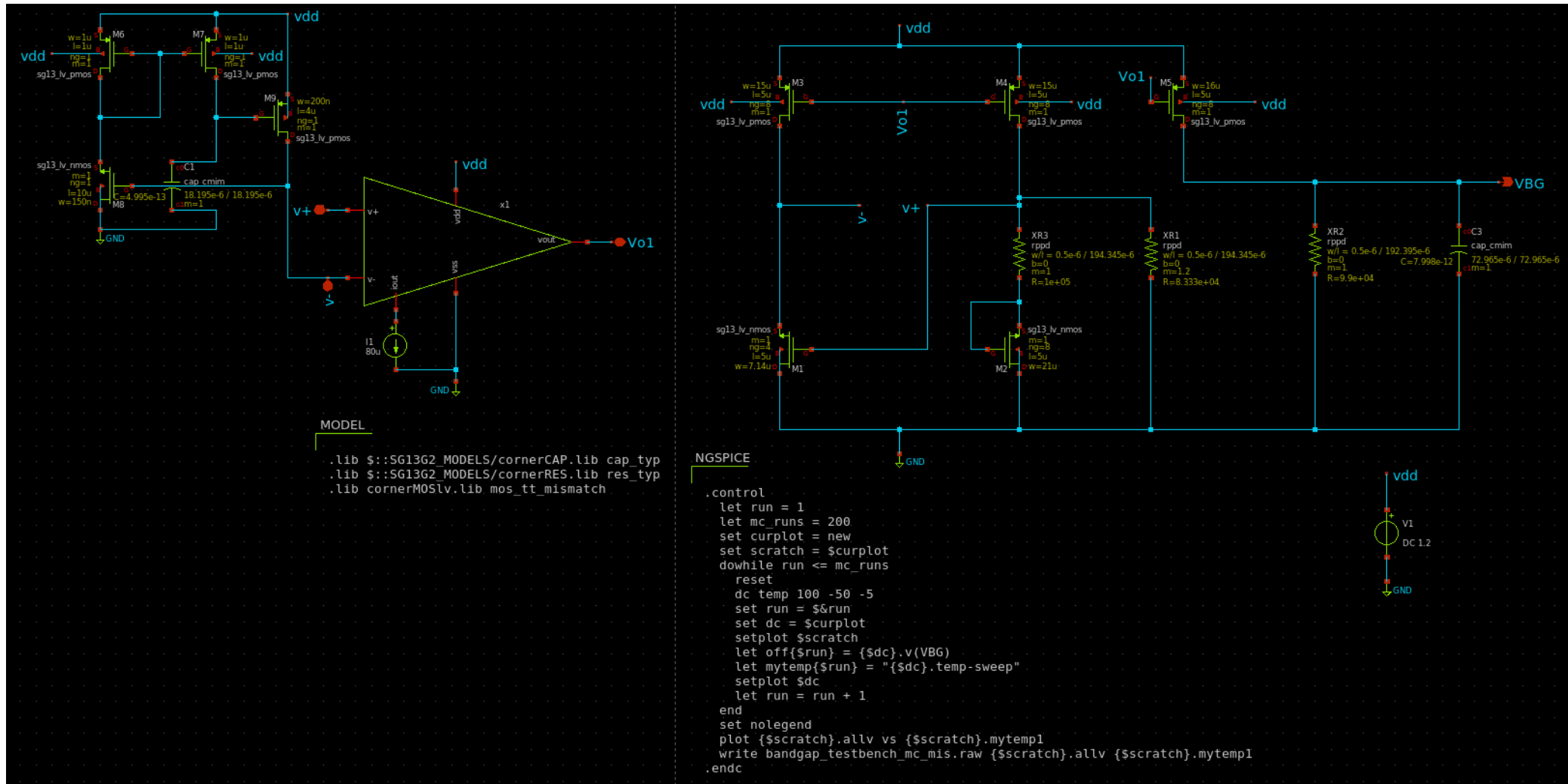
```
.param mm_ok=1
.param mc_ok=1
.control
  let run = 1
  let mc_runs = 100
  set curplot = new
  set scratch = $curplot
  dowhile run <= mc_runs
    reset
    dc temp 100 -50 -5
    set run = $&run
    set dc = $curplot
    setplot $scratch
    let off{$run} = {$dc}.v(VBG)
    let mytemp{$run} = \"{$dc}.temp-sweep\"
    setplot $dc
    let run = run + 1
  end
  set nolegend
  plot {$scratch}.allv vs {$scratch}.mytemp1
  write bandgap_testbench_mc_mis.raw {$scratch}.allv {$scratch}.mytemp1
.endc
```

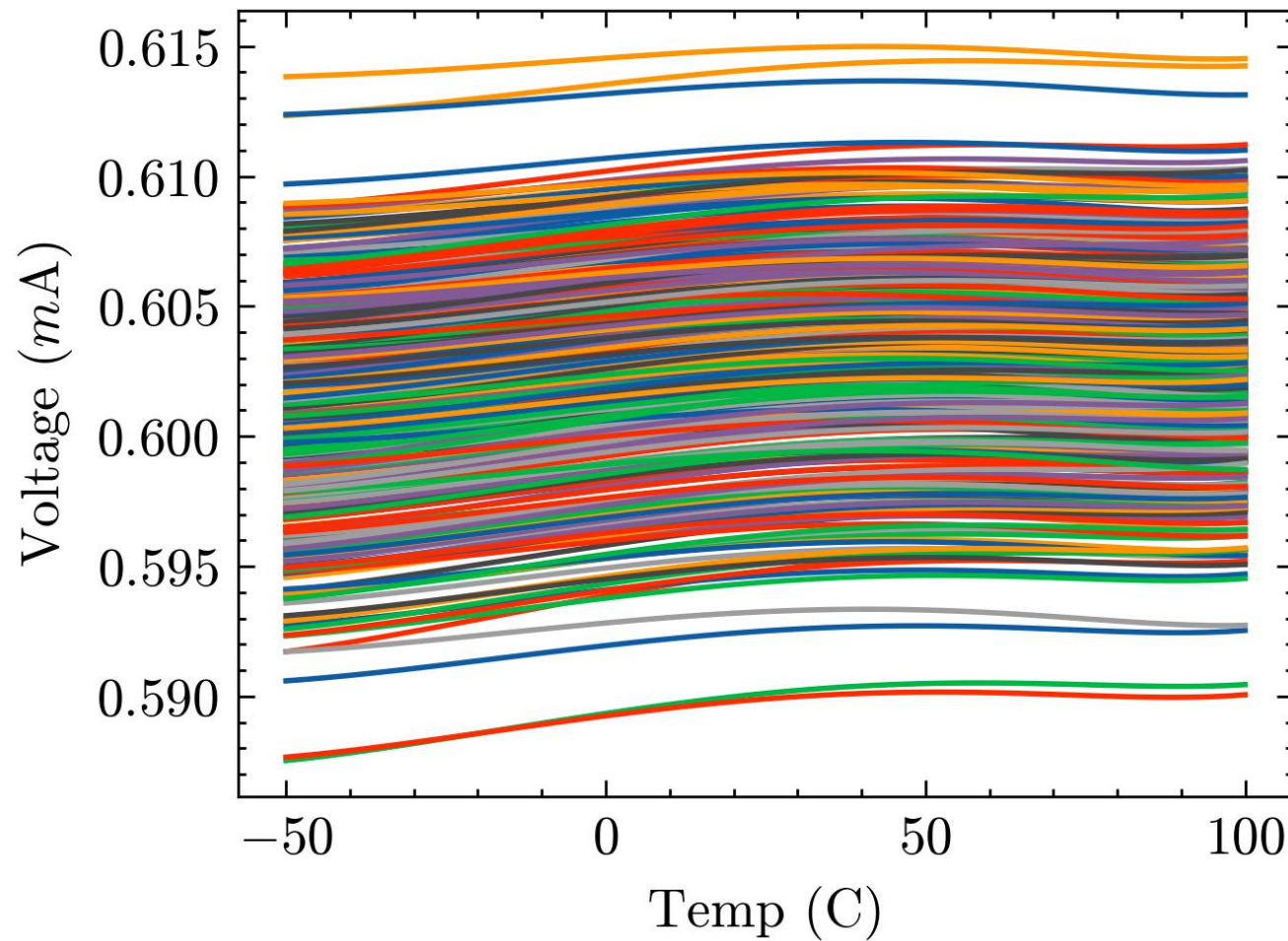
```
;Enable mismatch variation flag used by the model library
;Enable Monte Carlo variation flag used by the model library
;Begin ngspice control scripting block
;Initialize loop counter for Monte Carlo runs
;Define total number of Monte Carlo iterations
;Create a new plot to accumulate vectors from all runs
;Store the name of the accumulation plot for later access
;Start loop over all Monte Carlo iterations
;Reset simulator and regenerate random variations
;Run DC temperature sweep from 100 down to -50 in -5 steps
;Force numeric expansion of loop index variable
;Store current run's DC plot name
;Switch active plot to the accumulation plot
;Save VBG output voltage vector from this run into indexed result vector
;Save temperature sweep vector from this run into indexed result vector
;Switch active plot back to this run's DC plot
;Increment Monte Carlo run counter
;End Monte Carlo loop
;Disable legend in the plot output
;Plot all stored voltage vectors versus the first temperature vector
;Write all stored vectors and temperature axis to raw data file
;End ngspice control block
```

```
.lib $::SG13G2_MODELS/cornerCAP  
.lib cap_typ .lib $::SG13G2_MODELS/cornerRES  
.lib res_typ .lib cornerMOSlv  
.lib mos_tt_mismatch
```

Use .lib mos_tt_stat for statistical variation i.e wafer to wafer

Testbench







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Coffee Break!! / Catching Up



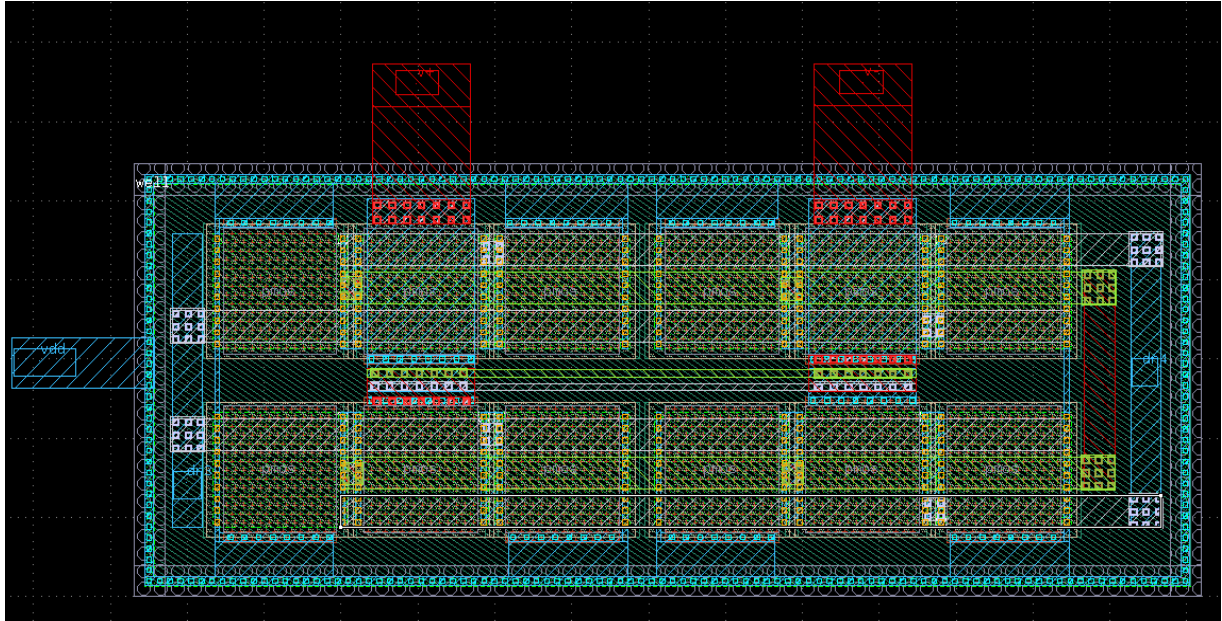
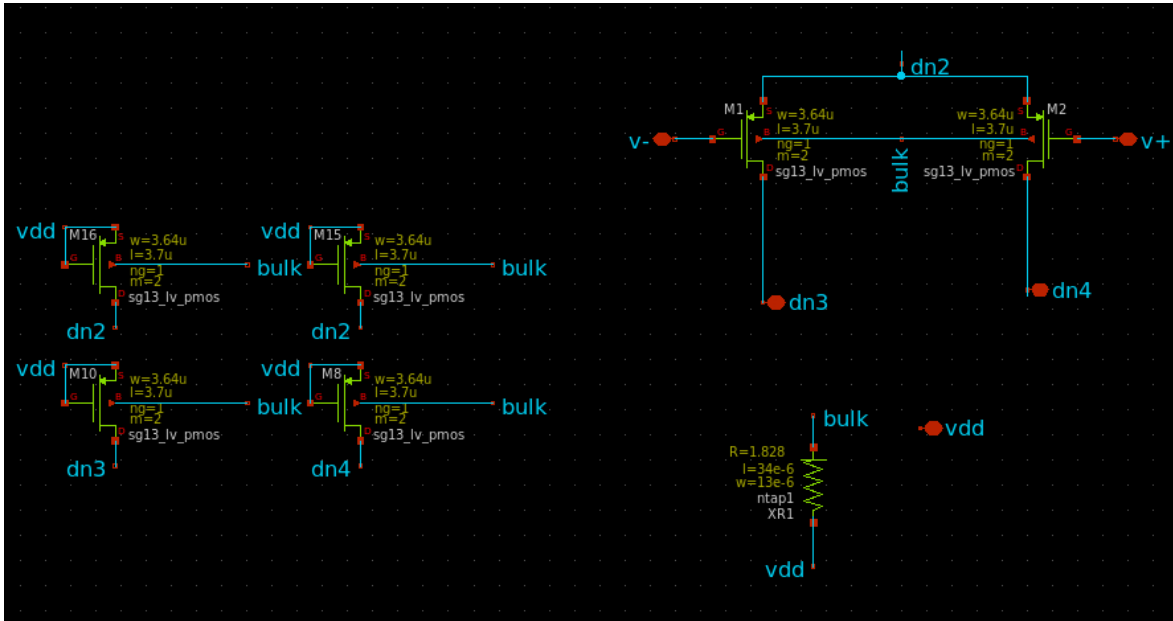
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Part 3

Layout

Reference: `/modules/module_1_bandgap_reference/part_3_layout`

Back Annotation



Objects	Layout	Reference
input_common_centroid ↔ INPUT_CO	input_common_centroid ⊖	INPUT_COMMON_CENTROID
▶ Pins ▶ Nets ▶ Devices		
▶ ntap1 ↔ NTAP1	⚠ \$13 / ntap1 [A=28.7556, P=0.18552k]	1 / NTAP1 [A=0.442k, P=94]

Back Annotation



```
8 *.PININFO v-:B v+:B vdd:B dn3:B dn4:B
9 M1 dn3 v- dn2 bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
10 M2 dn4 v+ dn2 bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
11 M8 dn4 vdd vdd bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
12 M10 dn3 vdd vdd bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
13 M15 dn2 vdd vdd bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
14 M16 dn2 vdd vdd bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
15 R1 vdd bulk ntap1 A=4.42e-10 P=9.4e-05
16 .ends
```

```
.subckt input_common_centroid v- v+ vdd dn3 dn4
*.PININFO v-:B v+:B vdd:B dn3:B dn4:B
M1 dn3 v- dn2 bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
M2 dn4 v+ dn2 bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
M8 dn4 vdd vdd bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
M10 dn3 vdd vdd bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
M15 dn2 vdd vdd bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
M16 dn2 vdd vdd bulk sg13_lv_pmos w=3.64u l=3.7u ng=1 m=2
R1 vdd bulk ntap1 A=28.7556e-12 P=0.18552e-03
.ends
```

Objects	Layout	Reference
input_common_centroid ⇌ INPUT_COM	input_common_centroid	INPUT_COMMON_CENTROID
↳ Pins		
↳ Nets		
↳ Devices		
↳ ntap1 ⇌ NTAP1	\$13 / ntap1 [A=28.7556, P=0.18552k]	1 / NTAP1 [A=28.7556, P=0.18552k]
↳ sg13_lv_pmos ⇌ SG13_LV_PMOS	\$4 / sg13_lv_pmos [L=3.7, W=7.28, AS=2.1]	1 / SG13_LV_PMOS [L=3.7, W=7.28]
↳ sg13_lv_pmos ⇌ SG13_LV_PMOS	\$3 / sg13_lv_pmos [L=3.7, W=7.28, AS=2.2]	2 / SG13_LV_PMOS [L=3.7, W=7.28]
↳ sg13_lv_pmos ⇌ SG13_LV_PMOS	\$5 / sg13_lv_pmos [L=3.7, W=7.28, AS=2.8]	8 / SG13_LV_PMOS [L=3.7, W=7.28]
↳ sg13_lv_pmos ⇌ SG13_LV_PMOS	\$6 / sg13_lv_pmos [L=3.7, W=7.28, AS=2.10]	10 / SG13_LV_PMOS [L=3.7, W=7.28]
↳ sg13_lv_pmos ⇌ SG13_LV_PMOS	\$1 / sg13_lv_pmos [L=3.7, W=14.56, AS=15]	15 / SG13_LV_PMOS [L=3.7, W=14.56]

Alternatively



- 0 Complete unfinished tasks from today or yesterday
- 0 Explore other design metrics for the OTA or Bandgap
- 0 Ask questions/ get help if something wasn't clear today
- 0 Relax 😊

What Will We Do Tomorrow?



- 0 **QUCS-S:** Getting familiar with QUCS-S and Xyce
- 0 **50GHz MPA:** Perform S-parameter analysis, along with DC and Harmonic Balancing
- 0 **Matching:** Performing simple matching using online tool and tuning in QUCS-S
- 0 **OpenEMS:** Expert talk on OpenEMS and Palace and how to interface with it using Python along with the new EMstudio
- 0 **EM Simulation:** EM simulating components for the 50GHz MPA